

- Introns (continued)
 exon shuffling and, 2:30–31
 function, 1:12–13, 2:52, 2:53, 3:181, 4:45, 4:110
 gene expression role, 2:62, 2:63
 in higher eukaryotes, 2:115, 4:128
 lacking in bacteria, 2:31, 4:128
 lacking in mitochondria and chloroplasts, 2:196, 3:56
 lacking in mRNA, 1:154, 1:157, 1:224, 2:53, 3:124, 3:211, 4:51–52, 4:57–59
 lacking in retropseudogenes, 4:10
 point mutations of, 2:27
 removal in transfer RNA, 4:59
 triplet repeats, 4:149, 4:150
See also Splicing, alternative; Splicing, RNA
- Inversions, and chromosomal aberrations, 1:122, 1:124–125, 1:124, 3:97
- IP-3, signal transduction role, 4:89–90
- IQ (intelligence quotient) tests
 Down syndrome, 1:79
 as eugenics tool, 2:18
 Flynn effect, 2:209
 FSIQ, 2:208
 history, 2:207–208
 MENSA, 2:208
 PIQ, 2:208
 Stanford-Binet Intelligence Scale, 2:208
 VIQ, 2:208
 Wechsler Adult Intelligence Scales, 2:208
- Iris
 origin of term, 2:32
 structure, and eye color, 2:31–33
- IS (insertion sequence) elements, prokaryotes, 2:116, 2:117, 4:144–145, 4:144
- Ishihara color vision test, 1:172
- Isoelectric focusing, 2:48, 3:208
- Isoforms, protein, 1:13, 1:14, 1:14, 2:31, 2:53, 3:160
- Isohaemagglutinin alleles, 3:82
- Isoleucine (Ile), genetic code, 2:85, 2:87, 4:137
- Isoleucine-valine operon (*ilv*), 3:134
- Isotretinoin (Accutane), 1:80–81
- IUCN (International World Conservation Union), categories of threatened populations, 1:186
- J**
- Jacob, Francois, 3:105, 3:131
- Jacobs, P.A., 1:119, 4:80
- JAK-STAT signaling pathway, 4:75–76, 4:77
- Janus kinase gene, 4:75–76
- Jefferson, Thomas, Y chromosome analysis, 3:65
- Jellyfish genes
 as bioremediation tool, 1:62
 escaped, 1:68
 green fluorescent protein, 3:17
- Jennings, Herbert Spencer, 2:20
- Jews
 evolutionary origin, 3:66
 Hasidic, founders effect, 2:37
 killed by Nazis, 2:20
- Jews, Ashkenazic
 breast cancer, 1:91
 cystic fibrosis, 1:202
 diabetes, 1:210
 idiopathic torsion dystonia, 2:38
 Tay-Sachs disease, 2:37, 2:99–100, 3:174, 4:98, 4:101–102
- Johansson, Wilhelm, 1:130, 2:50, 3:101
- Johnston, Stephen, 1:255
- Jost, Alfred, 4:79–80, 4:81
- Jumping genes. *See* Transposable genetic elements
- Juvenile-onset diabetes (diabetes type 1), 1:209–210, 2:202
- K**
- Kaback, Michael, 4:101
- The Kallikak Family* (H. Goddard), 2:18, 2:19
- Kallman, Franz, 4:83
- Kaposi's sarcoma, 2:151, 4:169
- Kappa* bacteria, as intracellular symbionts, 2:198
- Kappa constant regions, antibodies, 2:179, 2:179, 2:180
- Karyotype analysis
 to diagnose birth defects, 1:79
 human chromosomes, 1:127, 3:108
 spectral (chromosome painting), 3:183
 Turner's syndrome, 1:122
See also Chromosomal banding
- Karyotype, defined, 1:79
- Kaufman, Matthew, 4:126
- Kearns-Sayre syndrome, 3:53
- Kellogg's, genetically modified ingredients, 4:130
- Kendler, Kenneth, 4:83
- Kennedy disease (spinobulbar muscular atrophy), 1:26, 4:150, 4:151
- Khorana, Har Gobind, 1:193, 1:252–253
- Kidney disorders
 cancer, 1:93
 and diabetes, 1:209, 1:212
 metabolic diseases, 3:42
 mitochondrial diseases, 3:54
- Kiedrowski, Gunter von, 2:24
- Kilobases, defined, 3:7
- Kinase enzymes
 CAM/kinases, 4:90
 cyclin-dependent, 1:105–108
 defined, 1:105, 3:153, 4:38
 Janus, 4:75–76
 MAP, 4:90, 4:91
 phosphorylation by, 3:178, 4:87, 4:87
 PKA, 4:89
 PKC, 4:89
 pre-replicative complex enzymes, 4:18
 repetitive DNA sequences, 3:153–154
 RTK pathway, 4:87–88, 4:89, 4:90
 serine/threonine, 4:87, 4:89–90
 tyrosine, 4:87–89
- Kinetochore
 defined, 1:144
 structure and function, 1:114, 1:115, 3:25–26, 3:27, 3:28, 3:29, 3:58–59
- Kirk, Katherine, 4:83
- Klinefelter's syndrome
 cause and symptoms, 1:79, 1:121–122, 3:111, 4:80, 4:174
 progeroid aspects, 1:2
- Knight, Margaret, 1:15
- Knock-in mutants
 embryonic stem cell research, 2:5
 gene targeting, 2:73, 3:91
- Knock-out mutants
Arabidopsis transformation studies, 1:35
DAT gene, 1:41
 "Disease Model" mouse strains, 3:62
 embryonic stem cell research, 2:4–5
FMR-1 gene, 2:42
 gene targeting, 2:72, 2:73, 3:91, 4:54, 4:125–126
 Hox genes, 1:208
 pigs, for organ transplants, 4:140, 4:142–143
 PrP genes, 3:188
 SCID studies, 4:75
 technology for creating, 4:124

- Knowledge bases, bioinformatics tools, 1:56
- Knudson, Alfred Jr., 3:131
- Koch, Christof, 1:194
- Koonin, Eugene, 1:143
- Kosambi, Damodar, 3:13
- Kosambi map functions, 3:13
- Kossel, Albrecht, 1:249
- K-ras* gene, and colon cancer, 1:167–168
- Krebs cycle
- defects, 3:42, 3:43, 3:44
 - described, 1:111
- Kremer bodies. *See* PML bodies
- Kuru disease, cause, 3:189
- L**
- L929 cancer cells, 4:104–105
- La Quinta High School, tuberculosis, 1:26
- Laboratory information management systems, 1:54
- Laboratory technicians, 3:1–3, 3:2
- automated sequencing by, 1:43–45, 3:1
 - microbiologists as, 3:50–51
 - molecular biologists as, 3:70–72
 - plant genetic engineers as, 3:149–150
- Lac* operons
- discovery, 3:131
 - function, 2:15, 3:132
 - gene expression regulation, 3:132, 3:133, 3:134–135
- Lac Saint-Jean population, and Tay-Sachs disease, 4:101
- LacA* gene, 3:132
- Lactose, *lac* operons, 2:15, 3:131–135, 3:132, 3:133
- Lactose permease enzyme, 3:135
- LacY* gene, 3:132
- LacZ* gene, 3:132, 3:152
- Lahr, Marta, 3:168
- Lake Magadi (Kenya), Archaea adaptations, 1:37
- Lambda constant regions, antibodies, 2:179, 2:179
- Lambda phages, 4:119
- Lamin A/C* gene, and diabetes, 1:211
- Lamin proteins, 3:125
- Landsteiner, Karl, 1:82, 1:84, 1:233
- Laparoscopies, 4:21, 4:28
- Laron syndrome, 2:129
- Lassa fever virus, characteristics, 4:165
- Latanoprost, eye color changes and, 2:33
- Latent infections
- defined, 4:169
 - by viruses and retroviruses, 4:169
- Laughlin, Harry Hamilton, 2:19–20
- “Law for the Prevention of Genetically Diseased Progeny,” 2:19
- Law of independent assortment
- genes on homologous chromosomes, 3:36, 3:75
 - incomplete dominance, 3:36
 - linked genes, 3:5–6, 3:14–15
 - meiosis and, 3:26, 3:29, 3:60, 3:75
 - Mendel’s hypothesis, 1:130, 1:148, 3:31, 3:34–35
- Law of segregation, Mendel’s hypotheses, 1:130, 1:148, 3:31, 3:34
- Lawyers. *See* Attorneys
- Leading and lagging strands, DNA replication, 4:14, 4:14, 4:15–16, 4:15
- Learning disabilities, ADHD and, 1:39–40
- Learning, *psd-95* gene and, 2:5
- Leber’s hereditary optic neuropathy
- cause, 3:53
 - clinical features, 2:202
- Lederberg, Joshua, 1:141, 1:183–184
- Leewenhoek, Antoni van, 3:119
- Legal issues, 3:3–4
- antiscegenation laws, 2:17
 - blood typing, 1:233–234, 1:236
 - compulsory sterilization, 2:16–20, 2:90
 - cryopreservation, 4:26
 - DNA profiling, 1:73, 1:233–239, 3:3, 3:4
 - genetic discrimination, 1:68, 2:93–94, 3:3–4
 - genetically modified foods, 2:107, 3:3
 - genetics tools, 1:42
 - illegal trade of endangered species, 1:186, 1:189–190
 - inbreeding prohibitions, 2:189
 - patenting genes, 1:42–43, 1:72, 3:4, 3:136–138
 - right to privacy, 2:92, 3:4, 3:191–193
 - surrogacy, 4:24, 4:29
 - tissue ownership, 3:3–4
 - in vitro fertilization, 4:29
- Legionnaires’ disease, testing tools, 1:64
- Leigh, Dennis, 3:54
- Leigh’s disease, 3:52, 3:54
- Lejeune, Jerome, 1:119, 1:256
- Lentiviruses, 2:151, 4:35, 4:38
- Leprechaunism, 1:211
- Lesch-Nyhan syndrome, symptoms and treatment, 3:40, 3:44
- Lesions, defined, 1:167
- Leucine (Leu)
- chemical properties, 3:200
 - genetic code, 2:85, 2:87, 4:137
- Leukemia, 1:93, 1:94
- acute myeloid, 3:96, 4:11
 - adult T-cell, 4:34
 - chronic myelogenous, 3:127, 3:129, 4:154
 - human acute promyelocytic, 3:124
 - human Tlymphotropic virus, 4:34, 4:36, 4:38, 4:169
 - murine virus, 4:35, 4:36
 - T-cell, 4:38, 4:165, 4:169
 - tumor suppressor gene role, 4:154, 4:155
- Leukotrienes, as extranuclear hormones, 2:161
- Levene, Phoebus Aaron, 1:249–250
- Levine and Stetson, Rh system, 1:84
- Lexicon, functional genomics role, 2:124
- LGMD (limb-girdle muscular dystrophy), 3:86
- LH hormones, and endocrine disorders, 2:129
- LI* elements, 4:145–146
- Life span. *See* Aging and life span
- Lifestyle
- and cancer, 1:91, 1:92, 2:90
 - and cardiovascular disease, 1:6, 1:101, 2:105
 - and diabetes, 1:210
 - as disease treatment tool, 2:118
 - and longevity, 1:6
- Li-Fraumeni syndrome, 3:98, 4:154
- Ligands
- defined, 2:163, 4:87
 - GPCR pathway, 4:89
 - RTK pathway, 4:87–88
- Ligase enzymes, defined, 1:246, 4:33
- Ligation
- in BER pathway, 1:244
 - defined, 1:244, 3:158, 4:45
 - for DNA libraries, 1:223, 1:224
 - by ribozymes, 4:45
- Light and high-energy particles, as mutagens, 3:88
- Light chain polypeptides, 2:179, 2:179
- LINEs (long interspersed repeated elements), 4:8, 4:10–11, 4:144, 4:145, 4:147–148

- Linkage and recombination (linkage analysis), 3:4–8, 3:74
 as allelic scoring strategy, 2:192
 calculating linkage and map distance, 3:6
 CEPH families, 3:12, 3:13
 defined, 1:16, 4:3
 DNA microarray tools, 1:228
 as gene discovery tool, 2:59, 2:60
 genetic marker tools, 3:6, 3:11
 independent assortment role, 3:5–6, 3:75
 linkage disequilibrium mapping, 2:38
 lod score linkage, 3:8
 of maize, 3:10
 Morgan's studies, 3:7, 3:12, 3:14–15, 3:74, 3:75
 physical *vs.* genetic linkage, 3:5–6, 3:5
 Punnett and Bateson's contributions, 3:36
 statistical tools, 3:7–8, 3:12–13, 3:15
See also Crossing over (recombination)
- Linkage and recombination, specific diseases/traits
 Alzheimer's disease, 1:16–17
 bipolar disorder, 3:215
 cardiovascular disease, 1:101–103
 cystic fibrosis, 1:201
 hemophilia, 2:143
 Huntington's disease, 2:104–105
 Parkinson's disease, 4:159
 psychiatric disorders, 3:214–215
 sexual orientation, 4:84
- Linkage, defined, 3:4
- Linkage disequilibrium mapping, 2:38
- Lipid vesicles, as gene therapy vectors, 2:76–77
- Lipids
 in Archaeal membranes, 1:37
 and cardiovascular disease, 2:105
 defined, 2:11, 4:98
 ganglioside, 4:98–100
 in LPS layer, 2:11–12
 PIP-2, 4:89
 in viral envelope, 4:165
- Lipopolysaccharide (LPS) layer, 2:11–12
- Lipoprotein cholesterol, and cardiovascular disease, 1:101, 2:105, 3:43
- Liquid chromatography. *See* High-performance liquid chromatography
- Liver disorders
 cancer, 1:93
 Gaucher disease, 3:40, 3:45, 4:4
- Liver function
 diabetes and, 1:209
 poison detoxification, 1:45–46, 3:89
 promutagen activation, 1:20
- Livestock
 antibiotic resistance concerns, 1:28
 breeding, quantitative traits, 4:1
 cloning, 1:158, 1:160, 1:164
 cryopreservation, 4:25
 DNA vaccines, 1:254, 2:121
 genetically engineered, 1:11, 1:73, 2:125
 inbreeding, 2:146, 2:190
 pigs, for organ transplants, 4:140, 4:142–143
 prion diseases, 3:188–190
 selenium intoxication, 1:61
 transgenic animals, 2:106, 4:127, 4:130
- Loci, gene
 defined, 1:5, 2:7, 3:5, 4:1
 human, number, 2:7, 3:107
 marker, 3:11, 3:13–14
 physical *vs.* genetic linkage, 3:5–6, 3:5
 quantitative traits, 4:1–3
See also Mapping genes
- Locoweed (*Astragalus*), as bioremediator, 1:61
- Lod score linkage analysis, 3:8
- Long interspersed repeated elements (LINEs), 4:8, 4:10–11, 4:144, 4:145, 4:147–148
- Long terminal repeats (LTR), of proviruses, 4:36, 4:36, 4:37
- Longevity. *See* Aging and life span
- Longitudinal studies, population based, 1:103
- Longitudinally, defined, 1:114
- Looped domains
 of DNA, 1:126, 1:134, 1:134, 4:114
 gene expression role, 2:64, 2:65
- Los Alamos National Laboratory, 3:71
- Loss-of-function mutations
 albinism as, 2:32, 2:128, 2:200–201, 2:202, 2:206, 2:206
 defined, 2:74
 hemophilia as, 3:78, 3:97, 3:186
 inheritance patterns, 3:96
 metabolic diseases, 3:38–39
 muscular dystrophy as, 2:77–78, 2:80
- Lower Risk population status, 1:186
- LoxP sites, 2:73
- LPS (lipopolysaccharide) layer, 2:11–12
- LTR (long terminal repeats), of proviruses, 4:36, 4:36, 4:37
- Lubs, Herbert, 2:39
- Lucas, William, 3:38
- Lumen (endoplasmic reticulum)
 defined, 3:202
 translation role, 1:112–113
- Lung disorders
 cancer, 1:93, 1:180
See also Cystic fibrosis
- Lung transplants, 1:201
- Lungs, hemoglobin binding, 2:136
- Luria, Salvador, 1:204, 4:172
- Lyase reactions, of nucleases, 3:113
- Lykken, David, 2:9
- Lymphatic system
 cancer metastasis via, 1:94
 damaged by HIV/AIDS, 2:153
 lymphomas, 1:93, 1:94
- Lymphocytes
 cancerous tumors, 1:94
 cytotoxic T, 1:32
 defined, 1:3
 progeria and, 1:3
 transplant-rejection role, 4:140
- Lymphomas, 1:93, 1:94
 B-cell, 2:155, 3:146
- Lyon Hypothesis. *See* X chromosomes: inactivation of
- Lyon, Mary, 3:76–77, 4:176
- Lyonization. *See* X chromosomes: inactivation of
- Lyse, defined, 1:139
- Lysine (Lys)
 genetic code, 2:85, 4:137
 substitution, sickle-cell disease, 2:139
 ubiquitination, 3:179
- Lysis, defined, 4:168
- Lysogenic viruses, 4:118–119
- Lysosomes, 1:105
 ganglioside metabolism, 4:99
 storage disorders, 3:40–41, 3:44–45
- Lytic viruses, transduction cycle, 4:117–118, 4:118

M

- MacLeod, Colin, 3:103, 3:105, 4:124
- Macromolecular, defined, 1:239
- Macromolecules, defined, 2:186, 3:44
- Macronuclei, defined, 2:115

- Macrophages
 cancerous tumors, 1:94
 defined, 3:45
 function, 2:152, 2:178
 metabolic disorders, 3:45
- MACs (mammalian artificial chromosomes), 1:145, 1:146
- Mad cow disease. *See* Bovine spongiform encephalopathy (BSE)
- Magnesium ions
 in ribozymes, 4:45
 RNA polymerase role, 4:56
- Maize (corn), **3:8–11**
 crossing-over studies, 1:131, 3:21, 3:75
 genetic diversity, 3:9–10, 3:9
 genetically engineered, 1:9, 1:58, 1:68, 2:108, 3:149, 4:130, 4:134
 genome size, 1:34
 hybrid vigor, 2:146, 3:9
 mapping studies, 3:10
 as model organism, 3:8–9, 3:60, 3:149
 origin, 3:8
 production statistics, 3:8
 STARLINK, 1:58
 transposable genetic elements, 3:10, 3:21–22, 4:143–144
- Major and minor grooves (DNA), 1:217, 1:217, 1:218, 4:115, 4:116
- Major histocompatibility complex (human leukocyte antigen)
 in centenarians, 1:8
 and diabetes, 1:209–210
 function, 2:181, 2:183
 and transplant rejection, 2:181, 3:82, 4:139–140, 4:141
- Malaria
 causal organism, 2:148
 DNA vaccines, 1:255
 glucose-6-phosphate dehydrogenase deficiency and, 2:148
 sickle-cell disease and, 1:46, 1:203, 2:27, 2:135, 2:141, 2:148, 4:68
 testing tools, 1:64
 thalassemias and, 2:141, 2:148
- MALDI (matrix-assisted laser desorption ionization), 3:20, 3:208
- Malic acid, as phytochelatin, 1:60
- Malignant/malignancies
 defined, 1:97, 4:38
See also Cancer; Carcinomas; Tumors
- Mammals
 artificial chromosomes (MACs), 1:144–146, 1:145
 cloning, 1:158–160, 1:159, 1:163–164
 developmental processes, 1:204–206, 1:208–209
 DNA methylation, 3:46, 3:47
 exceptions to universal code, 2:87
 homeotic genes, 2:68
 Hox genes, 1:208
 hybrids, 1:146
 imprinting, 2:183, 2:185–186
 marker systems for, 3:16–17
 mutation rates, 1:240, 3:101
 nuclear pores, 3:125–126
 sex determination, 4:78–81
 transposable genetic elements, 4:145
- Mammograms, and breast cancer, 1:90, 3:175
- Manhattan Project, 2:172
- Manic depression, epistatic interactions, 2:9
- Manifesting carriers, 2:204
- Map functions
 defined, 3:12
 Haldane, 3:13
 Kosambi, 3:13
- MAP (mitogen activated protein) kinase cascades, 4:90, 4:91
- Maple syrup urine disease, symptoms and treatment, 3:40
- Mapping genes, **3:11–15**
 artificial chromosome tools, 1:145–146
 CEPH families, 3:12, 3:13
 clinical geneticist role, 1:149–151
 contigs, 2:60
 crossing-over frequency, 1:197, 2:59, 3:11–13, 3:26, 3:83
 disease identification role, 2:170, 3:11
 gene cloning tools, 1:152, 1:154–157, 2:11
 gene discovery role, 2:59, 2:60, 3:11
 genetic maps, 3:15
 genomics industry role, 2:123–125
 genomics role, 2:120–122
 Human Gene Mapping Workshop, 3:23
 interference, 1:197, 3:12–13
 linkage and recombination tools, 1:131, 1:141–142, 1:197, 2:59, 3:4–8, 3:26
 linkage disequilibrium mapping, 2:38
 molecular hybridization tools, 1:225–226
 physical maps, 1:155, 1:156, 3:7, 3:15
 polymorphic markers, 3:6, 3:7, 4:146
 repeated sequences as tools, 4:11
 restriction enzyme tools, 4:33
 restriction maps, 1:155, 1:156
in situ hybridization tools, 2:186–189
 SNP tools, 3:162
 transductional, 4:119–120
See also DNA sequencing; Human Genome Project
- Mapping genes, specific organisms
E. coli, 1:140, 1:141–142, 2:11, 2:86, 2:116
 fruit flies, 2:43, 3:7, 3:12–13, 3:14–15, 3:74, 3:75
 maize, 3:10
 roundworms, 4:62
- Marfan syndrome, 1:75, 1:76, 2:202
- Marker loci, 3:11, 3:13–14
- Marker systems, **3:15–18**
 how they work, 3:15–16
 screenable markers, 2:73, 2:188, 3:16, 3:17–18, 3:91
 selectable markers, 1:153, 2:72, 3:16–17, 3:152, 3:162
See also Genetic markers
- Maroteaux-Lamy syndrome, symptoms and treatment, 3:41
- Marriage, miscegenation laws, 2:16, 2:17
- Martha's Vineyard, hearing losses, 2:37
- Martin, J. Purdon, 2:39
- Mass spectrometry, **3:18–21**
 applications, 3:20, 3:198
 procedures, 3:18–20, 3:19
 as proteomics tool, 3:206, 3:208
 in quantitative trait studies, 4:3
- Maternal and Child Health Bureau (Children's Bureau), PKU screening, 3:219
- Maternal serum screening tests, 2:97, 3:185
- Maternally inherited diabetes and deafness (MIDD), 1:211
- Matrix-assisted laser desorption ionization (MALDI), 3:20, 3:208
- Matthaei, Johann, 1:252–253
- Maturity onset diabetes (MODY), 1:209, 1:211
- Maxam-Gilbert sequencing, 4:71
- McCarty, Maclyn, 3:103, 3:105, 4:124
- McClintock, Barbara, 3:21–22, 3:21
 breakage-fusion-bridge cycle, 3:21, 3:22
 crossing-over studies, 1:131, 3:10, 3:75, 3:105

- McClintock, Barbara (continued)
transposable genetic elements, 3:10, 3:21–22, 3:97, 4:143–144
- McKusick, Victor, 3:10, 3:22–24, 3:23
- MCM (mini-chromosome maintenance) proteins, 4:18
- MD. *See* Muscular dystrophy
- MDR (multi-drug resistance), 1:26, 1:27, 1:28
- Measles, German (rubella)
and birth defects, 1:75, 1:81
vaccine, 4:129
virus, characteristics, 4:165
- Measured (bottom-up) genotype strategy, 4:3
- The Mechanism of Mendelian Heredity*, 3:76
- Meconium ileus, 1:201
- MeCP2 protein, and Rett syndrome, 3:48, 3:49, 3:78
- Media (for bacterial cultures), defined, 3:50, 4:22
- Medical genetics. *See* Clinical geneticists
- Medical records, privacy of, 3:191–192
- Mediterranean heritage
sickle-cell disease, 2:137, 2:148
thalassemias, 2:140, 2:148
- Medium-chain acyl-CoA dehydrogenase deficiency, newborn screening, 3:176
- MegaBase sequencer, 1:44–45
- Megan the sheep, 1:163
- Meiosis, 3:24–30
centromere role, 1:114–115, 3:25, 3:29
chiasmata role, 1:194, 1:195, 3:26
chromosomal aberrations, 1:79, 3:29
chromosome packaging errors, 1:257
cytokinesis, 3:29
defined, 1:194, 2:35, 3:5, 4:68
distinguished from mitosis, 3:24, 3:26, 3:29, 3:59, 3:60
early studies, 1:130, 1:149
gamete formation, 1:115, 2:33–34, 3:24–25, 3:75
independent assortment of chromosomes, 3:26, 3:29
kinetochore role, 3:25, 3:26, 3:27, 3:28, 3:29
microtubule role, 3:26, 3:27, 3:29
purpose, 3:24
spindle role, 1:135, 3:26, 3:111–112
- See also* Chromosomes, eukaryotic; Crossing over (recombination); Nondisjunction
- Meiosis I, 3:24–26
anaphase, 3:26, 3:27, 3:28
gamete dormancy, 2:34–35, 3:29
metaphase, 3:26, 3:27, 3:28
prophase, 3:25–26
telophase, 3:26
- Meiosis II, 3:25, 3:29
gamete dormancy, 2:35, 3:29
metaphase, 3:27, 3:28
telophase, 3:27
- Melanin
and albinism, 2:32, 2:128, 2:200–201, 2:206
in iris, 2:32
- Melanocytes, defined, 2:32
- MELAS (mitochondrial encephalopathy, lactic acidosis, and stroke), 2:202, 3:54
- Melatonin, as extranuclear hormone, 2:161
- Membrane proteins, structure, 3:201
- Memory loss
aging and, 1:7
Alzheimer's disease and, 1:7, 1:14–15, 1:180
psd-95 gene and, 2:5
- MEN1* gene, and Multiple endocrine neoplasia, 4:153
- Mendel, Gregor, 3:30–32, 3:30
- Mendelian genetics, 3:32–37
character pairs, 3:101–102
current status, 2:111
dihybrid crosses, 3:35
eugenics applications, 2:16, 2:18
exceptions to, 3:35–36, 3:75
gene identification, 1:155
law of independent assortment, 1:130, 1:148, 3:5–6, 3:14–15, 3:26, 3:29, 3:31, 3:34–35, 3:60, 3:75
law of segregation, 1:130, 1:148, 3:31, 3:34
- The Mechanism of Mendelian Heredity*, 3:76
- meiosis and, 1:115
microsatellite inheritance, 1:189
monohybrid crosses, 3:33
procedures, 3:30–31, 3:32–34
Punnett squares, 1:148, 3:33, 3:35
trihybrid crosses, 3:35
- See also* Chromosomal theory of inheritance, history; Classical hybrid genetics; Gene nature, history; Inheritance patterns; Simple (Mendelian) traits
- Mendelian Inheritance in Man* (V. McKusick), 3:23
- Mendelism. *See* Mendelian genetics
- Mennonites, inbreeding studies, 2:190–191
- Menopause, aging and, 1:7
- MENSA, 2:208
- Menstruation
AIS individuals, 1:25
apoptosis role, 1:32
and breast cancer risk, 1:90
normal development, 1:23
ovulation and, 2:34–35, 3:29
in vitro fertilization and, 4:21
- Mental retardation
eugenics to eliminate, 2:17–20
genetic components, 3:213
intelligence as inherited trait, 3:81
newborn screening to prevent, 2:98–99, 3:42–43
- See also* Down syndrome; Fragile X syndrome; Intelligence; Phenylketonuria
- Merck, drug development, 2:124
- Mercuric reductase, 1:62
- Mercury, hyperaccumulators of, 1:62
- Meselson, Matthew, 1:251–252
- Messenger RNA. *See* RNA, messenger (mRNA)
- Metabolic diseases, 3:37–46
amino acid metabolism disorders, 3:40
birth defects, 1:75
cofactor or vitamin enhancement, 3:39
detoxifying agents, 3:39, 3:45
diet modifications to treat, 3:39, 3:43, 3:44, 3:45
enzyme defects as cause, 3:38–39, 3:39
enzyme replacement therapy, 3:39, 3:45
fatty acid metabolism disorders, 3:40
gene therapy, 3:39
inheritance patterns, 3:38–39
organ transplants, 3:39, 3:45
organic acid metabolism disorders, 3:40
peroxisomal metabolism disorders, 3:41
prenatal genetic testing, 3:183–184, 3:186
purine and pyrimidine metabolism disorders, 3:40, 3:40–41
urea formation disorders, 3:41
- Metabolism
aerobic, 2:195

- anabolic reactions, 1:110
 ATP and, 1:110–111, 2:194, 3:51–52, 3:55–56, 3:117
 by-products, DNA damage, 1:239
 catabolic reactions, 1:110, 4:99
 defined, 1:78, 1:110, 3:37
 enzymatically controlled reactions, 3:37–38
 of gangliosides, 4:99–100
 hormonal regulation, 2:160
 inborn errors of, 3:37
 mitochondrial role, 1:111–112, 2:116, 2:194, 3:41, 3:55–56
 normal *vs.* impaired gene sequences, 3:39
 signal transduction role, 4:87
 Metabolites, defined, 3:142, 3:218
 Metageria, progeroid aspects, 1:2
 Metallothioneins, 1:60
 Metals (heavy)
 bioremediation to remove, 1:60
 plasmid resistance to, 3:151
 Metamorphosis, apoptosis and, 1:32
 Metaphase
 chromosomal banding techniques for, 1:126
 I, meiosis, 3:26, 3:27, 3:28
 II, meiosis, 3:27, 3:28
 mitosis, 3:59
 prometaphase, mitosis, 3:59
 Metaphase plate, 3:59
 Metastasis, defined, 1:94, 1:227
 Metazoans, axis development, 2:68
 Methane
 produced by Archaea, 1:37
 produced by eubacteria, 2:13
Methanobacterium thermoautotrophicum, genome characteristics, 2:121
 Methionine (Met)
 genetic code, 2:84, 2:85, 2:87, 4:137
 oxidation of, 1:63–64
 as start codon, 2:84, 4:4, 4:137, 4:138
 Methylation, of DNA, 3:46–49
 CpG islands, 3:47, 3:48
 defined, 2:185, 4:31
 gene imprinting, 3:48
 gene repression role, 3:48, 3:77, 3:100
 host defense function, 3:46, 3:47–48
 and human disease, 3:49
 imprinting role, 2:184, 2:185, 2:185
 mutations and, 3:95, 3:100
 process, 3:46–47, 3:47
 restriction enzyme role, 3:47, 4:31
 Methylation, of proteins, 3:203, 3:205, 4:111
 Methyl-binding proteins, gene repression role, 3:48
 Methylmalonic acidemia, symptoms and treatment, 3:40, 3:43
 Methylmalonyl-CoA, 3:43
 MHC proteins. *See* Major histocompatibility complex
 Mice. *See* Rodent models
 Microbiologists, 3:50–51
 Microcephaly, defined, 1:80
 Microliters, defined, 3:156
 Micrometers, defined, 1:108
 Micronuclei, defined, 2:115
 Micronucleus test, 3:92–93
 Microsatellites. *See* Short tandem repeats
 Microscopes
 electron, to study viruses, 4:164–165
 invention of, 1:130
 Microtubules
 defined, 1:144
 structure and function, 3:26, 3:27, 3:29, 3:59, 3:121
Microviridae, genome characteristics, 2:117
 Miescher, Johann Friedrich, 1:249
 Migration. *See* Immigration and migration
 Millennium Pharmaceuticals, gene mining by, 2:123
 Miller, Stanley, 2:23
 Miniature inverted-repeat transposable elements (MITEs), 4:9
 Mini-chromosome maintenance (MCM) proteins, 4:18
 Minimal-gene-set concept, 1:143
 Mining, bioremediation tools, 1:60
 Minisatellites. *See* Variable number of tandem repeats
 Minor histocompatibility genes, 4:140
 Miscegenation
 defined, 2:17
 eugenics and, 2:16, 2:17
 Mismatch distributions (pairwise differences), 3:167–168, 3:168
 Missense mutations, 2:127, 3:153
 MITEs (miniature inverted-repeat transposable elements), 4:9
 Mitochondria, 1:105, 1:111
 defined, 1:33, 2:194, 3:135, 4:42
 diseases, MIDD, 1:211
 electron transport chain, 1:111, 2:194, 3:41–42, 3:51–53, 3:52
 and oxidative base damage, 1:242
 plasmids in, 3:150
 ribosomes, 4:42
 ribozymes, 4:44
 in sperm, 2:35
 structure and function, 1:111–112, 3:41–42, 3:51, 3:55–56, 3:56, 4:50
 Mitochondrial diseases, 3:51–55
 aging and, 3:80
 genetic components, 3:52–53
 inheritance of, 2:194, 2:198, 2:204
 Kearns-Sayre syndrome, 3:53, 3:53
 Leber's hereditary optic neuropathy, 3:53
 Leigh's disease, 3:52, 3:54
 MELAS, 2:202, 3:54
 mosaicism, 3:79–80
 multisystem, list of, 3:54
 oxidative metabolism disorders, 3:41–42
 Mitochondrial DNA. *See* DNA, mitochondrial (mtDNA)
 Mitochondrial encephalopathy, lactic acidosis, and stroke (MELAS), 2:202, 3:54
 Mitochondrial Eve, 3:63, 3:67–68
 Mitochondrial genome, 3:55–57
 base pairs, number, 2:116, 3:79
 coding for ETC subunits, 3:52–53, 3:52
 distinguished from eukaryotes, 2:195–196
 evolutionary origin, 2:12, 2:86, 2:112, 2:116, 2:194–196, 3:52, 3:55, 3:56–57
 as evolutionary study tool, 2:198, 3:66–67
 exceptions to universal code, 2:86–87
 heteroplasmic and homoplasmic, 2:197, 3:54
 inheritance patterns, 2:195, 2:196–197, 2:204, 2:205, 3:51, 3:67–68, 3:69, 4:157
 mutations, 3:42, 3:79–80
 overlapping genes, 3:135–136
 random partitioning, 2:195, 2:197
 random replication, 2:195, 2:197, 3:56
 repair mechanisms lacking, 3:79
 vegetative segregation, 2:197
 See also DNA, mitochondrial (mtDNA)

- Mitogen activated protein (MAP) kinase cascades, 4:90, 4:91
- Mitosis, 3:57–60, 3:58
 anaphase, 3:59
 cell cycle phases, 1:104, 1:104, 3:57–59
 centromere role, 1:114, 3:58
 cytokinesis, 3:59
 defined, 3:106, 4:152
 distinguished from meiosis, 3:24, 3:26, 3:29, 3:59, 3:60
 kinetochore role, 3:58–59
 metaphase, 3:59
 prophase, 3:59
 spontaneous tetraploids, 3:166
 telophase, 3:59
 tumor suppressor gene role, 4:162
 vegetative segregation, 2:197
See also Chromosomes, eukaryotic; Chromosomes, prokaryotic; DNA replication
- Mob* genes, 3:152
- Mobile elements. *See* Transposable genetic elements
- Model organisms, 3:60–62
 Archaea, 1:38
 behavior genetics, 1:48–49
 developmental processes, 1:204–205
 disease identification studies, 2:170
 examples, 3:60–61
 genome sequencing, 2:120–121, 2:172, 2:173
 homologous chromosomes of, 2:122
 pigs, 4:141, 4:142–143
 useful traits and attributes, 1:33–34, 2:10, 2:43–44, 3:60–61, 3:73
See also Arabidopsis thaliana; *Escherichia coli*; Fruit flies; Maize (corn); Rodent models; Roundworms; Yeasts
- Modified Gomori trichome stains, 3:53, 3:53
- Modifier genes, Muller's concept, 3:81
- MODY (maturity onset diabetes), 1:209, 1:211
- Molecular anthropology. *See* Anthropology, molecular
- Molecular biologist. *See* Biologists: molecular
- Molecular Biology of the Gene* (J. Watson), 4:173
- Molecular clocks
 divergence from common ancestry, 3:63, 3:94, 3:98, 3:100–101, 3:167–168
 precision, 3:67
- Molecular cloning. *See* Cloning genes
- Molecular Dynamics, automated sequencers, 1:44–45
- Molecular systematics
 defined, 2:158
See also Taxonomy (systematics)
- Molecules, evolution. *See* Evolution, molecular
- Monochromat color vision defects, 1:171, 1:172
- Monocots, genetic modification techniques, 2:107–108
- Monocytes, cancerous tumors, 1:94
- Monod, Jacques, 3:131
- Monohybrid crosses, 3:33
- Monomers, defined, 3:178
- Monosomy
 defined, 1:121
 impact of, 1:121, 1:123, 3:109, 3:111
- Monozygotic, defined, 4:83
- Monozygotic twins. *See* Twins, identical
- Montagneur, Luc, 4:40
- Montmorillonite, and ribose-phosphate backbone, 2:23
- Moore v. Regents of the University of California*, cell line ownership, 3:3–4
- Moorhead, P. S., 4:105
- Morag the sheep, 1:163
- Morgan, Thomas Hunt, 3:72–76, 3:73
 collaboration with Delbrück, 1:204
 fruit fly genome mapping, 2:42–43, 3:7, 3:12, 3:14–15, 3:74, 3:75
 genes described by, 1:155, 1:249, 2:50, 3:32, 3:73
 X-linked inheritance studies, 1:130–131, 3:73–74, 3:102
- Morgan, unit of measurement, 3:75
- Morphogens
 defined, 1:206
 gradients, and development, 1:206
- Morphologically, defined, 4:62
- Morphological-species concept, 4:91–92
- Morquio syndrome, symptoms and treatment, 3:41
- Morton, Newton, 2:148
- Mosaicism, 1:120, 3:76–80
 calico cat example, 3:77, 3:78
 cause and symptoms, 1:123
 chromosomal, 3:79
- Down syndrome, 1:257
 germinal, of progeria, 1:3
 inheritance patterns, 2:204
 mitochondrial, 3:79–80
 nonrandom inactivation, 3:78
 X-chromosome inactivation, 3:77–78, 4:175–176
- Mouse mammary tumor virus, 4:35
- MRNA. *See* RNA, messenger
- Mu opioid receptors, 1:5
- Mucoid, defined, 4:123
- Mucopolysaccharidosis, symptoms and treatment, 3:45
- Mucous membranes, defined, 4:134
- Muller, Hermann, 3:80–81, 3:80
The Mechanism of Mendelian Heredity, 3:76
 mutation rate measurements, 3:80
 on polyploidy, 3:164
 radiation-induced mutations, 3:81, 3:88, 3:103
 Watson and, 4:172
 X-linked inheritance studies, 1:130–131, 2:42–43, 3:73, 3:91–92
- Müllerian ducts, 1:22, 1:23, 4:79, 4:79
- Müllerian inhibiting substance, 4:79
- Mullis, Kary, 2:172, 3:91, 3:154
- Multi-drug resistance (MDR), 1:26, 1:27, 1:28
- Multifactorial inheritance
 about, 1:77, 1:178
 threshold model, 1:77, 1:77
- Multifactorial traits
 defined, 2:206
 inheritance patterns, 2:206
- Multimers, defined, 1:146, 3:202, 4:45
- Multinucleate, defined, 2:115
- Multiple alleles. *See* Alleles, multiple
- Multiple endocrine neoplasia, 4:153
- Multiple epiphyseal dysplasia, 2:132
- Multiple genetic hit theory, 3:131, 4:153
- Multiple sclerosis, family studies, 2:168–169
- Multiplicative effects of genes, 1:178–179, 1:179
- MuM phage, 4:144
- Municipal wastes, bioremediation tools, 1:60
- Munitions dumps, bioremediation tools, 1:62
- Murine, 4:35–36
- Murine leukemia virus, 4:35, 4:36
- Muscle and skeletal disorders, birth defects, 1:75

- Muscle disorders
 Kearns-Sayre syndrome, 3:53, 3:53
 mitochondrial diseases, 3:54
 roundworms to study, 4:63
 spinal muscular atrophy, 3:124
See also Muscular dystrophy
- Muscles
 actin proteins, 2:29
 cell shape, 1:109
 gene therapy, 2:78
 healthy tissue, 3:84
 MyoD protein, 2:65
 myoglobin proteins, 2:29
- Muscular dystrophy (MD), 3:83–87
 Becker (BMD), 3:84
 characteristics, 3:83
 drug therapies, 3:87
 Emery-Dreifuss, 3:125
 gene therapy, 2:75, 2:77–78, 2:80, 2:81, 3:86–87
 limb-girdle (LGMD), 3:86
 prenatal genetic testing, 3:186
- Muscular dystrophy, Duchenne (DMD)
 causative gene, 3:84–85, 3:100
 characteristics, 3:84, 3:85
 gene therapy, 2:73
 genetic testing, 2:168, 3:85, 3:184
 inheritance patterns, 1:213, 2:191–192, 2:202, 2:203, 3:78, 3:83–84
 mutation rates, 3:100
- Muscular dystrophy, myotonic
 causative gene, 3:86
 characteristics, 3:85–86, 4:150
 genetic testing, 3:86
 inheritance patterns, 3:85, 4:150
 pleiotropic effects, 3:153–154
 as triplet repeat disease, 2:42, 3:86, 4:11, 4:150
- Mustard gas, as mutagen, 3:90
- Mutable genes, 3:21
- Mutagenesis, 3:89–93
 defined, 4:182
 induced, research role, 3:89, 3:90–92, 3:97–98
 site-directed, 3:91
 in zebrafish, 4:182–183
See also Carcinogenesis
- Mutagens, 3:87–89
 Ames test, 1:19–21, 1:20, 1:100, 3:87, 3:92
 base analogs, 3:87, 3:88, 3:90–91
 base-altering, 3:88
 defined, 1:19, 3:89
 distinguished from carcinogens, 3:87
 impact on tumor suppressor genes, 4:152–153
 and increased mutation rates, 3:99–100
 intercalating agents, 3:88
 promutagens, 1:20
See also Carcinogens; Teratogens
- Mutagens, specific
 aflatoxins, 1:244–245
 asbestos, 3:89
 benzo[*a*]pyrene, 1:244–245, 1:245, 3:89
 5-bromo-deoxyuridine, 3:87
 dimethylsulfate, 1:242
 ethidium bromide, 3:88
 free radicals, 3:88, 3:89
 industrial toxins, 1:77, 1:119, 1:123, 1:210, 1:214, 1:239, 1:242
 light and high-energy particles, 3:88
 mustard gas, 3:90
 nitrite preservatives, 3:88
 smoke, 1:77, 1:92, 1:98, 1:100, 1:179–180, 1:242, 3:88, 3:89
 transposons, 3:91, 3:97
See also Radiation, ionizing; Radiation, ultraviolet
- Mutation analysis
 as gene discovery tool, 2:59
 pedigrees as tools, 3:140
- Mutation fixation, 1:241–242, 1:241
- Mutation rates, 3:98–101
 amino acids, 3:94
 breast cancer genes, 1:91–92
 deamination errors, 1:241
 DNA repair gene, 1:242
 Duchenne muscular dystrophy, 3:100
 eggs *vs.* sperm, 2:192
 factors that influence, 3:99–100
 hemophilia, 2:143
 HIV, 2:155, 2:157
 human genome, 3:99, 3:177
 Huntington's disease, 3:100
 hydrolysis errors, 1:240
 mammals, 1:240, 3:101
 measuring, 3:80, 3:99
 molecular clocks, 3:63, 3:67, 3:98, 3:100–101
 point mutations, 2:27
 reduced by antioxidants, 3:89, 3:100
- Mutations, 3:93–98
 alkylation errors, 1:240, 1:242
 apoptosis studies, 1:32–33
 back, 3:65, 3:100
 and cell cycle regulation, 1:107–108
 cloning implications, 1:160
 and complex traits, 1:177–181
 conservative, 2:157
 de novo, 1:120
 deamination errors, 1:240–242, 1:240
 defined, 1:92, 2:21, 3:63, 4:17
 distinguished from polymorphisms, 2:126
 distinguished from SNPs, 2:118
 and divergence from common ancestry, 3:63, 3:65, 3:167–168, 4:178–179
 DNA libraries of, 2:122
 and DNA repair function, 1:230
 enzyme defects, 3:38
 exogenous and endogenous, 1:242
 in extranuclear genes, 2:194, 2:198
 gene expression role, 2:54, 2:65–66
 genetic analysis role, 2:43
 in germ-line cells, 3:93, 3:99
 in Hardy-Weinberg model, 2:134
 hydrolysis errors, 1:240, 1:240
 impact on protein structure, 1:53
 of maize, 3:9–10, 3:9
 missense, 2:127, 3:153
 in mitochondrial DNA, 3:42, 3:51, 3:53–54
 Muller's hypotheses, 3:80–81
 neutral, overlapping genes and, 3:136
 in noncoding regions, 3:65, 3:82–83, 3:93, 3:99
 nonsense, 2:127, 3:153
 organism's tolerance of, 2:157–158, 4:67–68
 outcrossing and, 1:35
 oxidation errors, 1:240, 1:242
 phenotype manifestations, 1:35, 1:75, 1:155, 1:156–157
 premutations, 2:40–42
 in pseudogenes, 2:29–30, 3:209–212
 rarity of, 3:90
 reverse, 1:19–20
 selective advantages, 3:94
 silent, 2:127, 2:157, 3:93, 3:96, 3:99
 of sperm, 3:99, 4:25
 splice junction, 4:101
 spontaneous, 1:97, 3:90, 3:100, 4:75, 4:153
 transition, 3:95

- Mutations (continued)
 transversion, 3:95
 of viral genes, 2:128
See also Chromosomal aberrations; DNA repair; Frameshift mutations; Gain-of-function mutations; Knock-in mutants; Knock-out mutants; Loss-of-function mutations; Mosaicism; Pleiotropy; Point mutations; *specific diseases*
- Mutations, detecting
 Ames test, 1:19–20, 1:20–21, 1:20, 3:92
 artificial chromosome study tools, 1:145
 chromosomal aberrations, 3:92–93
 homologous sequence tools, 2:157–158
 and mutation rates, 3:100
 restriction and physical maps, 1:155–156, 1:156
 somatic, 3:93
- Mutations, specific genes
 ;gDF508, 1:200, 1:202
 A3243G, 3:54
 ADA, 4:75
 androgen receptor gene, 1:21, 1:23–25, 1:24
antennapedia, 1:207–208, 1:208, 2:65–66
 APC, 1:167–168, 4:146, 4:153
 BRCA1 and BRCA2, 4:153
 CDKN1C, 4:153
c-myc, 3:96
 COMP, 2:132
 DNMT3B, 3:49
 DRD4, 1:41
dystrophin, 3:85, 3:100
 factor VIII, 2:143, 4:174
 fibroblast growth factor receptor, 2:67, 2:130
 FIX, 2:143, 2:145
 FMR-1, 3:49
 GH, 2:129
 globin proteins, 2:137
 HEXA and HEXB genes, 4:98, 4:99, 4:99, 4:100–101
hprt, 3:93
 IL-2 cytokine receptor, 4:75
 MEN1, 4:153
neo^r gene, 4:125–126
 NF1 and NF2, 3:130, 3:153, 4:153
p53, 2:5, 3:130, 4:154–155, 4:169
 phenylalanine hydroxylase gene, 2:55
 proto-oncogenes, 1:99–100, 1:155, 3:127–130
 PrP, 3:188–189, 3:190
psd-95, 2:5
 RAG1 and RAG2, 4:76
 Rb, 1:96–97, 1:96, 3:130, 4:153
 TK, 4:125–126
 TSC1 and TSC2, 4:153
v-erbB, 3:130
v-fms, 3:130
- Mycobacteria, inteins, 3:181
Mycobacterium tuberculosis
 antibiotic resistance, 1:26
 genome characteristics, 1:142, 2:121
Mycoplasma, antibiotic resistance, 1:27
Mycoplasma capricolum, exceptions to universal code, 2:87
Mycoplasma genitalium, genome characteristics, 1:142, 1:143, 2:116, 2:172
Mycoplasma pneumoniae, genome characteristics, 1:142
- Myeloid blood cells, cancerous tumors, 1:94
- Myelomas, defined, 1:93
- MyoD protein, forced expression of, 2:65
- Myoglobin proteins
 defined, 2:29
 evolutionary origin, 3:212, 3:212
 tertiary structure, 3:201
- Myosin
 function, 3:199
 in roundworms, 4:63
- Myotonic dystrophy
 clinical features, 2:202
 progeroid aspects, 1:2
- Myrmecia pilosula*, chromosome number, 2:113
- Myxococcus xanthus*, chromosome size, 2:116
- N**
- Na⁺/K⁺ ATPase pump, 3:204
- Nagasaki (Japan), radiation-caused mutations, 2:171, 2:173
- Nagel's anomaloscope color vision test, 1:172
- Nanometers, defined, 1:133, 1:170
- Nash family, preimplantation genetic diagnosis, 3:186, 4:29–30
- Nathans, Daniel, 1:71
- National Academy of Sciences, Human Genome Project role, 2:173
- National Center for Biotechnology Information (NCBI), 1:142, 2:156, 2:176, 2:212
- National Institutes of Health (NIH)
 genetic discrimination studies, 2:94
 Human Genome Project role, 2:173–174, 2:176
 population screening, 3:177
- National Research Council, Human Genome Project role, 2:173
- National Science Foundation, Human Genome Project role, 2:173
- National Society of Genetic Counselors, 2:90, 2:91
- Native Americans
 diabetes prevalence, 1:210
 eugenic fitness ranking, 2:17
- Natronobacterium*, 1:37
- Natronococcus*, 1:37
- Natronosomonas*, 1:37
- Natural enemies, as biopesticides, 1:57–58
- Natural law ethics, 4:28
- Natural selection
 for antibiotic-resistant bacteria, 4:44
 distinguished from genetic drift, 2:70–71
 and divergence from common ancestry, 3:63
 duplicate genes, removal/conservation of, 2:28, 2:69, 2:158
 eugenics and, 2:17
 genetic basis for, 1:131, 2:21, 2:26, 2:29
 in Hardy-Weinberg model, 2:134, 2:135
 intron removal and, 2:31
 mutations, consequences, 3:93–94, 4:67–68
 mutations, removal of harmful, 2:158
 speciation and, 2:156, 4:92
- Nature of the gene. *See* Gene nature
- Nazi Germany, eugenics movement, 2:16, 2:90, 4:26
- NCBI (National Center for Biotechnology Information), 1:142, 2:156, 2:176, 2:212
- ND10. *See* PML bodies
- Neandertals
 distinguished from modern humans, 3:66–67, 3:66, 3:69, 3:167
 DNA analysis, 3:68–69
 replaced by modern humans, 3:170

- Necrosis, defined, 1:31, 4:169
- Negative selection marker systems, 3:16
- Neisseria gonorrhea*, antibiotic resistance, 1:26, 1:27, 1:28
- Neisseria meningitidis*, genome characteristics, 1:142
- Nematodes
- as biopesticides, 1:57
 - defined, 1:57, 4:62
 - holocentric chromosomes, 1:114
 - plant resistance to, 1:149
 - See also Roundworms
- Neonatal, defined, 1:3
- Neoplasms
- defined, 1:92
 - See also Tumors
- Neo^r* gene, targeting studies, 4:125–126
- NER (nucleotide excision repair), 1:244–246, 1:245, 1:247
- Neural tube defects (NTDs)
- as multifactorial disorder, 1:77–78
 - prenatal genetic testing, 3:184, 3:185
- NeuroD1/beta2* genes, and diabetes, 1:211
- Neurofibromatosis
- Alu sequences and, 4:146
 - genomic screening, 2:168
 - pleiotropic effects, 3:153, 3:154
 - tumor suppressor genes, 4:153
- Neuroimaging, defined, 1:40
- Neurological crises, metabolic diseases, 3:42
- Neurological, defined, 3:42
- Neurological disorders
- mitochondrial diseases, 3:54
 - Parkinson's disease, 1:228, 2:5–6, 3:190, 4:159, 4:160–161
 - polyglutamine disorders, 4:151
- Neurons
- defined, 1:15, 3:45
 - FMR-1 protein and, 1:76
 - membrane transport, 1:109
- Neurospora*, biochemical genetics studies, 3:103
- Neurotransmitters
- defined, 1:5, 4:85
 - for dopamine and serotonin, 1:5, 1:39–41, 1:41
 - signal transduction role, 4:85
- New Synthesis, 3:32
- Newborn screening, 2:98–99
- congenital adrenal hyperplasia, 3:176, 3:177
 - cystic fibrosis, 3:177
 - galactosemia, 3:44
 - hemoglobinopathies, 3:177
 - hypothyroidism, 3:176–177
 - medium-chain acyl-CoA dehydrogenase deficiency, 3:176
 - phenylketonuria, 3:42–43, 3:176, 3:218–219
- Newborn screening tests
- DNA- vs. non-DNA based, 3:176–177
 - Guthrie, 3:176
 - hemoglobin electrophoresis, 3:176
- Neyman, Jerzy, 4:97
- NF1* and *NF2* genes, 3:130, 3:153, 4:153
- Nic sites, on DNA, 3:151, 3:151
- Nickel, hyperaccumulators of, 1:61
- Nicotine addiction. See Smoking
- NIH (National Institutes of Health)
- genetic discrimination studies, 2:94
 - Human Genome Project role, 2:173–174, 2:176
- Nirenberg, Marshall, 1:193, 1:252–253
- Nitrite preservatives, as mutagens, 3:88
- Nitrogen fixation, transgenic plants, 2:106
- Nobel Prizes
- bacterial genetics, 1:183–184
 - blood group systems, 1:82
 - chromosomal theory of inheritance, 3:76, 3:80
 - DNA sequencing methods, 2:172, 4:64–65
 - DNA structure, 2:171, 4:172
 - Kuru transmission, 3:189
 - multiple, Fred Sanger, 4:64
 - oncogene research, 4:38
 - operon discovery, 3:131
 - PCR invention, 2:172, 3:91, 3:154
 - phage genetics, 1:204
 - prion hypothesis, 3:187
 - radiation-induced mutations, 3:81
 - recombinant DNA, 1:71, 1:72
 - restriction enzymes, 1:71
 - retroviral reverse transcriptase, 4:35, 4:40
 - RNA catalysts, 4:44
 - site-directed mutagenesis, 3:91
 - transposable genetic elements, 3:10, 3:22
- Nomenclature, 3:106–108
- Noncoding DNA sequences. See Introns
- Nondisjunction, 1:120, 3:108–112
- aging and, 3:112
 - and aneuploidy, 1:257, 3:110–111, 3:166
 - and chromosomal mosaicism, 3:79
 - defined, 1:121
 - early studies, 1:131
 - fatal vs. non-fatal conditions, 3:111
 - mechanism, 3:108–109, 3:109
 - spindle checkpoint errors, 3:111–112
- Non-insulin-dependent diabetes mellitus (diabetes type 2), 1:209, 1:210, 1:212, 3:125, 3:154
- Nonpolar, defined, 1:116, 3:200
- Nonprocessed pseudogenes, 3:210–211, 3:210
- Nonsense mutations, 2:127, 3:153
- Nontemplate strands, 4:107
- NOR-staining of chromosomes, 1:128
- Northern blotting
- defined, 1:86
 - procedure, 1:86–88
- Novartis, drug development, 2:124
- NPY mice, 1:5
- NTDs (neural tube defects)
- as multifactorial disorder, 1:77–78
 - prenatal genetic testing, 3:184, 3:185
- N-terminus, of amino acids/proteins, 3:181, 3:197–198, 3:207
- Nuclear hormone receptor superfamily
- anabolic steroids, 2:163
 - function, 2:161–163
 - mutations, consequences, 4:115
 - number, in humans, 2:161
 - number, in roundworms, 2:161–162
 - RXR partners, 2:160
- Nuclear lamina, 3:120, 3:121, 3:121, 3:125
- Nuclear localization signals, 3:126
- Nuclear magnetic resonance, 3:71, 4:48
- Nuclear membranes (envelopes), 1:105
- breakdown and reformation, 3:59, 3:120
 - defined, 1:139
 - structure and function, 1:112, 1:139, 3:120, 3:120, 3:122, 3:123, 3:125
- Nuclear pores, structure and function, 3:120, 3:121, 3:122–123, 3:123, 3:125–126

- Nuclear transplantation
 amphibian cloning, 1:161–163
 as gene therapy tool, 2:73
 mammal cloning, 1:162, 1:163–164
 transgenic animals, 4:124–126
See also In vitro fertilization
- Nuclease enzymes, 3:112–115
 crossing-over role, 1:195, 1:196
 defined, 1:31, 4:53
 deoxyribonucleases, 3:112, 3:113–114
 DNase, 1:201, 1:220, 3:221
 endonucleases, 3:112–113, 3:113, 3:114–115
 exonucleases, 3:112–113, 3:113–115, 3:113
 function, 1:31, 3:112
 hydrolysis reactions, 3:113, 3:114
 lyase reactions, 3:113
 ribonucleases, 3:112, 3:114–115
 structure preferences, 3:112–113
See also Restriction enzymes
- Nucleic acid
 defined, 2:49, 3:187
 early studies, 1:249, 2:21, 3:103, 3:104, 3:105
 electrophoresis to separate, 2:49
- Nucleocapsids, HIV, 2:152
- Nucleoids
 defined, 1:139, 2:14
E. coli, 1:141
 structure and function, 2:14–15, 2:112
- Nucleolar organizer, 3:123–124
- Nucleoli, 1:105, 1:111
- Nucleolus
 defined, 4:58
 ribosome synthesis, 4:43, 4:58
 structure and function, 3:120, 3:121, 3:123–124
- Nucleomorphs, overlapping genes, 3:136
- Nucleoplasm
 defined, 3:121
 structure and function, 3:121, 3:122–123
- Nucleoporins, 3:125–126
- Nucleoside analogues, 4:171
- Nucleoside triphosphates, defined, 1:230
- Nucleosides
 defined, 4:41
 and DNA structure, 1:215–220
 structure, 3:115, 3:117
- Nucleosomes, 1:134
 defined, 4:111
 structure and function, 1:126, 1:133–134, 4:111, 4:114
- Nucleotide analogs
 azidothymidine, 2:155–156
 ganciclovir, 3:17
- Nucleotide excision repair (NER), 1:244–246, 1:245, 1:247
- Nucleotides, 3:115–119
 antisense, 1:29–31, 1:30, 1:97, 2:106
 defined, 1:13, 2:22, 3:82, 4:9
 and DNA structure, 1:216–220, 2:50, 2:51, 3:94, 3:116, 4:12
 function, 3:115
 number in human genome, 1:73, 4:7, 4:13
 proportions, Chargaff's studies, 1:250–251, 4:172
 radioactive labeling, 1:220
 in RNA, 4:46–47, 4:47, 4:48, 4:49, 4:54, 4:109, 4:109, 4:135
 RNA, evolution of, 2:22–25
 structure, 3:115, 3:116, 3:117
See also Base pairs; DNA sequencing; Genetic code; RNA sequencing
- Nucleus, 1:111, 3:119–126, 3:121
 Cajal bodies, 3:120, 3:124, 4:50
 centrosomes, 3:121
 chromatin, 3:120, 3:120, 3:121
 chromosomal territories, 3:121–122, 3:122
 discovery, 3:119
 gems, 3:124
 interchromatin compartment, 3:120, 3:121, 3:122–123, 3:122
 intermediate filaments, 3:120, 3:121
 microtubules, 3:121
 nuclear lamina, 3:120, 3:121, 3:121, 3:125
 nuclear pores, 3:121
 nucleolus, 3:120, 3:121, 3:123–124
 nucleoplasm, 3:121, 3:122–123
 origin of term, 3:119
 PML bodies, 3:124–125
 polyglutamine aggregates in, 4:151–152
 speckles, 3:120, 3:125
 structure, 105, 1:112, 1:132–133, 2:112, 3:119–121, 3:120, 3:121, 4:12
 techniques for studying, 3:121, 3:122, 3:124, 3:125
See also Endoplasmic reticulum; Ribosomes
- Nutrition (diet)
 amino acid requirements, 3:41
 antioxidants in, 3:89
 and birth defects, 1:78
 and cancer, 1:92
 and cancer of breast, 1:91
 and colon cancer, 1:170
 as disease treatment tool, 2:118
 of early humans, 3:63, 3:68–69
 familial hypercholesterolemia and, 2:57
 and hormone regulation, 2:164
 and longevity, 1:6
 and phenylketonuria, 2:55–56, 2:99, 2:119, 3:176
 to treat metabolic diseases, 3:39, 3:43, 3:44, 3:45
- O**
- Obesity
 and cardiovascular disease, 1:101
 and diabetes, 1:210
 “Disease Model” mouse strains, 3:62
- O'Brien, Chloe, 3:186
- O'Brien, Stephen J., 1:189
- Obsessive-compulsive disorder (OCD), 3:213, 3:215
- Occipital cortex, 1:170
- Odorant receptors, signal transduction role, 4:89
- OI (osteogenesis imperfecta), 2:132
- Oil spills, bioremediation, 1:59, 1:60, 1:61, 1:62, 4:129
- Oils, genetically engineered, 1:10
- Okazaki fragments, 4:14, 4:15, 4:16, 4:17, 4:18
- Oligogenic traits, 1:178
- Oligonucleotide synthesizers, 1:176–177, 3:155
- Oligonucleotides, antisense, 4:183
- Omenn syndrome, 4:76
- Omphalocele, defined, 2:132
- Oncogenes, 3:127–131
 and cancer growth, 1:95, 1:99–100
 and cell cycle regulation, 1:107–108, 4:169
 defined, 1:99, 4:34
 as dominant, 3:130
 early research, 3:127
 growth-signaling pathways and, 3:128, 3:129–130
 interactions with environment, 1:180
 multiple genetic hits, 3:131
 mutations and, 3:98
 proto-oncogenes, 3:127, 3:128–129, 4:152
 retroviral role, 4:34, 4:38

- in situ* hybridization tools, 3:129
transcription factors as, 3:128, 3:130
- Oncogenes, specific
breast cancer, 1:91–92
cellular (*c-onc*), 4:38
colon cancer, 1:167–170
Ras, 1:99–100, 3:128, 3:130
v-erbB, 3:130
v-fms, 3:130
viral (*v-onc*), 4:38
v-sis, 3:130
- Oncogenesis, defined, 1:92
- Oncoproteins, viral, 4:155
- Oocytes
age of, 4:25
collecting, assisted reproduction, 4:21
defined, 1:121, 2:45, 3:24, 4:21
forced maturation of, 4:25
frozen, 4:25
fruit fly development, 1:206–207, 1:207, 2:45
primary, 3:24, 3:29
- Open reading frames (ORFs)
of cDNA, 1:154
defined, 2:53, 4:145
of DNA transposons, 4:9, 4:145
of pseudogenes, 3:209
- Operons, 3:131–135
discovery, 3:131
function, 1:142, 2:15, 3:105
functional relationships, 3:132
gene clustering, 3:132
lac, 2:15, 3:131–135, 3:132, 3:133
operator regions, 3:132, 3:134
promoter regions, 3:132, 3:134
regulation mechanisms, 3:134–135
transcription of, 3:134–135
- Opbioglossum reticulatum*, polyploidy, 2:114
- Opportunistic agents (pathogens), 2:151, 2:155
- Opsin proteins
role in color vision, 1:171
signal transduction role, 4:89
- Oracle databases, 1:55
- ORC (origin recognition complex), 4:13, 4:13, 4:18
- ORFs. *See* Open reading frames
- Organelles, defined, 2:108, 3:79, 4:42
- Organic acid metabolism, disorders, 3:40
- Organic, defined, 3:88
- Orgel, Leslie, 2:24
- OriC point, prokaryote chromosomes, 1:141
- Origin of replication (*ori*) sequences, 1:152–153, 3:151, 3:151, 4:12–13, 4:13
- Origin recognition complex (ORC), 4:13, 4:13, 4:18
- Ornithine transcarbamylase deficiency, 1:69, 3:45
- Orthologs, 2:158
- Oryza sativa*. *See* Rice
- Osmotic, defined, 2:11
- Ossification
defined, 2:130
endochondral, 2:130
- Osteoarthritis, 2:131–132
defined, 2:131
- Osteogenesis imperfecta (OI), 2:132, 2:201
- Osteoporosis
aging and, 1:7
defined, 1:1, 2:90
genetic counseling for, 2:90
progeria and, 1:1
- Ostreococcus tauri*, size, 1:108
- Outcrossing
Arabidopsis thaliana, 1:35
vs. inbreeding, fitness, 2:146
of maize, 3:9
Mendel's studies, 1:147
- Ovalbumin, function, 3:199
- Ovarian hyperstimulation syndrome, 4:22
- Ovaries
cancer, 1:93, 1:169, 2:102, 3:218, 4:22
development, 1:21, 1:22, 4:78–79, 4:79, 4:80–81
function, 2:34, 2:160, 3:24, 3:60
removal, and breast cancer, 1:90
tumors, and breast cancer, 1:91
- Overdominance hypothesis, for heterozygote advantage, 2:147
- Overlapping genes, 2:85–86, 3:135–136
- Oviduct, defined, 1:163
- Ovulation
defined, 1:163
drugs to enhance, 4:21, 4:22, 4:24, 4:28
and meiosis, 2:34–35, 3:29
pregnancy role, 4:19
- Ovules
defined, 1:35
and Mendelian ratios, 1:130
- Ovum
defined, 1:160
personhood of, 4:27–28, 4:29
- Oxidation
defined, 1:63
errors, DNA damage, 1:240, 1:242
of methionine, 1:63–64
- Oxidative metabolism, 3:41–42
- Oxidative phosphorylation
ATP production, 3:52
by mitochondria, 3:51–52, 3:55–56, 3:57
reactive oxygen species, 3:52
- Oxygen
consumed by mitochondria, 2:194
as electron acceptor, 2:13, 3:41–42
globin carriers, 2:136, 2:137–138
reactive species, as mutagens, 3:52, 3:100
- Oxytocin
and maternal condition, 4:29
post-translational control, 3:181
- P**
- P values, in statistical analyses, 3:194–196, 4:97
- P22 virus, 4:118
- P53* gene
and cancer, 1:167–168, 2:5, 4:154–155, 4:169
mutations, 1:95, 3:98, 3:130
- Pac* (puromycin-n-acetyl-transferase) gene, as selectable marker, 2:72
- PAGE (polyacrylamide gel electrophoresis), 2:46–47
2-D, 2:48–49, 2:48, 3:207–208
SDS, 2:47–48
- PAH (phenylalanine hydroxylase), 1:82, 2:55, 2:99, 3:42–43, 3:176, 4:5
- PAHs (polycyclic aromatic hydrocarbons), 1:59, 1:99
- Pairwise differences, in DNA sequences, 3:167–168, 3:168
- PAIS (partial androgen insensitivity syndrome), 1:25
- PAI-1 (plasminogen activating inhibitor 1), 1:8
- Palindromic DNA sequences, 1:218
- Palister-Hall syndrome, 2:130
- Pancreas, insulin production, 1:209
- Pancreatic disorders
cancer, 1:93
cystic fibrosis, 1:200, 1:201
mitochondrial diseases, 3:54
- Pancreatic elastase, 2:27
- Pandemics, defined, 1:28
- Pangenesis, 1:130

- Panic disorder, genetic components, 3:213
- Pap smears, 3:175
- Papaya, genetically engineered, 2:109
- Papaya ringspot potyvirus, 1:58
- Paracentric inversions, 1:123
- Paracrine hormones, 2:160
- Paralogous genes and pseudogenes, 3:210
- Paralogs, 2:158
- Paramecium* sp.
C value paradox, 2:114
intracellular symbionts of, 2:198
ploidy, 2:115
- Parental genomic imprinting, 2:183
- Parkinson's disease
and cigarette smoking, 4:161
DNA microarray analysis, 1:228
protein conformation role, 3:190
stem cell research tools, 2:5–6
symptoms, 4:160
twin studies, 4:159, 4:160–161
- Particle-mediated transformation, 4:132–133
- Partitioning, random
of mitochondrial and chloroplast genomes, 2:195, 2:197
of plasmids, 2:198
- Parvovirus, characteristics, 4:165
- The Passing of the Great Race* (M. Grant), 2:19
- Patent agents and attorneys, 1:42–43, 3:138
- Patent depositories, 3:138
- Patenting genes, 3:136–138
history, 1:72–73, 3:136–137, 4:129
Human Genome Project, 2:174, 2:175–176
international protection, 3:138
legal issues, 3:3–4
procedure, 3:138
requirements, 3:137–138
- Patents, defined, 3:136
- Paternity
anonymous, assisted reproduction, 4:28
DNA profiling to identify, 1:238
- Pathogenicity islands, 1:27–28
- Pathogenic/pathogenicity
defined, 1:27, 2:124, 3:151, 4:134
virulence plasmids, 3:151
- Pathogens
changes, immune system response to, 2:178
defined, 1:57, 2:10, 4:34
opportunistic, 2:151, 2:155
- See also* Eubacteria; Prions; Retroviruses; *specific pathogens*; Viruses
- Pathology/pathological, defined, 1:16, 1:171, 4:94
- Pathophysiology, defined, 1:39
- Pauling, Linus, 3:95, 3:200
- PBR322, cloning vector, 4:128
- PCBs (polychlorinated biphenyls), bioremediation to remove, 1:59
- PCR. *See* Polymerase chain reaction
- Peach latent mosaic viroid, 4:163
- Pearson, Karl, 3:102, 4:95
- Peas (garden), Mendelian genetics, 1:146–148, 3:30–31, 3:32–36
- Pedigrees, 3:138–142
confidentiality of, 3:142
defined, 3:7
degrees of relationship, 3:141
distinguished from genograms, 3:139
drawing and recording, 3:140–141
of ethnic backgrounds, 3:141–142
in Mendelian genetics, 3:36
sample, 3:140
symbols, 3:139, 3:140, 3:141
terminology, 3:140
uses, 3:139–140
Y chromosomes as tools, 3:65, 4:178
- Penetrance/nonpenetrance
age-dependent, 1:18
epistatic interactions, 2:8
incomplete, 2:201, 2:205
reduced, 2:61, 3:214
- Penicillin, resistance to, 1:26, 1:27, 1:185, 2:11, 4:69
- Peptide bonds
defined, 4:137
and protein structure, 3:197, 3:200, 4:138
ribosomal RNA role, 4:42, 4:137
- Peptide mass fingerprinting, 3:208
- Peptide sequencing. *See* Protein sequencing
- Peptides
defined, 3:20, 4:85
fusion, 4:167
- Peptidoglycan, in cell walls, 1:36, 2:11
- Peptidyl (P) site, in translation, 4:138
- Peptidyl-transferase center, 4:138
- Percutaneous, defined, 2:97
- Percutaneous umbilical blood sampling, prenatal genetic testing, 2:97–98
- Pericentric inversions, 1:123–124
- Perkin-Elmer sequencers, 2:173
- Peroxisome metabolism, disorders, 3:41, 3:45–46
- Peroxisomes, 1:105
- Perutz, Max, 4:151
- Pesticides
biopesticides, 1:57–59
damage to natural enemies, 1:57
resistance to, 1:11
role in disease, 1:214
- Pets
cloned, 1:158–159, 1:159
illegal trade in, 1:186
- Pfam database, 3:209
- PG (polygalacturonase) gene, 1:29
- PGD. *See* Preimplantation genetic diagnosis
- The phage group, 1:204
- Phagocytic, defined, 1:33
- Pharmaceutical scientists, 3:142–144, 3:145
- Pharmacogenetics and pharmacogenomics, 3:144–147
antisense nucleotide tools, 1:29, 1:30–31
apoptosis studies, 1:33
Arabidopsis research, 1:35
bacterial genome mapping studies, 1:142
bioinformatics tools, 1:53–54, 1:55–56
biopharmaceuticals, 2:125
cardiovascular disease treatments, 1:103
cost and access concerns, 1:69
DNA microarray tools, 3:146
gene cloning tools, 1:153–154, 1:163–164
genetically engineered products, 1:10–11, 1:63–64, 1:72, 1:74, 4:5–7
genomics industry role, 2:124–125
Human Genome Project contributions, 2:121–122, 2:171, 3:146–147
pharmaceutical scientist role, 3:142–144
predisposition predictions, 3:146, 3:148
side effects, preventing, 2:118–120, 2:122, 3:144–146, 4:22
signal transduction targets, 4:91
SNP comparisons, 3:146–147
transgenic animal models, 2:106, 4:127
See also Drugs (medications); Genomic medicine

- Phenocopies, defined, 2:61
- Phenotypes
- Bombay, 2:8, 2:9
 - defined, 1:21, 2:8, 3:9, 4:1
 - endophenotypes, 4:2, 4:2
 - Hardy-Weinberg predictions from, 2:135
 - mutation manifestations, 1:35, 1:75, 1:155, 1:156–157, 2:43
 - QTL effects on, 4:2–3
 - simple *vs.* complex, 1:177, 2:55, 2:60–61
 - See also* Genotype, phenotype and; *specific diseases and traits*
- Phenylalanine (Phe)
- genetic code, 2:85, 4:137
 - and PKU, 3:42–43
- Phenylalanine hydroxylase (PAH), 1:82, 2:55, 2:99, 3:42–43, 3:176, 4:5
- Phenylketonuria (PKU)
- as frameshift mutation, 4:5
 - gene-environment interactions, 2:55–56
 - Guthrie test, 3:176
 - maternal condition and, 1:82
 - newborn screening, 2:99, 2:119, 3:42–43, 3:176, 3:218–219
 - statistics, 1:75
 - symptoms and treatment, 2:202, 3:40, 3:42–43, 3:176
- Phenylthio-carbamide (PTC), ability to taste, 2:191
- Pheromones, defined, 2:158
- Philadelphia chromosome, 4:154
- Phimthop, Thawiphop, 1:78
- PhiX174 bacteriophage, 3:135, 4:64
- Phosphatase enzymes
- in DNA sequencing, 4:70
 - post-translational phosphorylation by, 3:178
 - signal transduction role, 4:87
- Phosphate groups
- defined, 1:105, 4:70
 - and DNA structure, 1:105, 2:50, 2:51, 3:115, 3:116, 3:117
- Phosphatidylcholine, function, 1:242
- Phosphodiester bonds
- defined, 1:244, 3:112
 - and DNA structure, 1:216, 1:216, 1:244, 3:117–118, 3:118
 - hydrolysis of, 3:113, 3:114
 - lyase reactions, 3:113
 - and RNA structure, 4:47, 4:49
- Phosphoglycerate kinase 2, 2:28
- Phospholipase C, signal transduction role, 4:89
- Phospholipids, in cell membranes, 1:109, 2:12, 3:55
- Phosphorimidazolid, and ribose-phosphate backbone, 2:23, 2:24
- Phosphorous, toxicity, 1:61
- Phosphorylation
- of antibiotics, 3:17
 - by ATP, 3:178, 3:204, 4:87
 - by CREB, 4:89
 - defined, 1:53, 2:62, 3:51, 4:86
 - DNA replication initiation role, 4:18
 - by GDP and GTP, 4:88, 4:89–90
 - of proteins, 1:105–107, 2:62, 3:20, 3:178, 3:180, 3:203–204, 3:205, 4:86–87, 4:87, 4:111
 - of resistance genes, 3:16–17
 - of RNA polymerase, 4:113
 - RTK pathway, 4:87–88
 - signal transduction role, 4:86–87, 4:87
 - See also* Kinase enzymes
- Phosphorimidazole, nucleotide linking role, 2:24
- Photopigments, role in color vision, 1:170–172
- Photoproducts, as mutagens, 1:244
- Photoreceptors
- Arabidopsis* research, 1:36
 - role in vision, 1:170–172
- Photosynthesis
- by chloroplast ancestor, 2:195
 - chloroplast role, 1:112, 2:108, 2:194
- Phototrophs, described, 2:13
- Phylogenetic Species Concept, 1:189
- Phylogenetic/phylogeneticists, defined, 1:36, 4:41, 4:146
- Physical maps (genes), 1:155, 1:156
- Physician, defined, 3:147
- Physician scientists, 3:147–148
- Phytochelutins, 1:60
- Phytophthora infestans*, potatoes resistant to, 2:108
- Phytoremediation, 1:60–61
- Picornaviruses, structure, 4:165
- Pigs
- cloned, 4:143
 - knock-out, for organ transplants, 4:140, 4:142–143
- Pilus, sex, 1:183, 1:183, 1:184, 2:15, 3:151
- Pineapples, genetically engineered, 2:107
- PIP-2 lipids, signal transduction role, 4:89
- PIQ (performance ability) intelligence scores, 2:208
- Pituitary glands
- and endocrine disorders, 2:129
 - function, 2:160
 - suppressing, assisted reproduction, 4:21
- PKA regulatory II_β mice, 1:5
- PKA (protein kinase), signal transduction role, 4:89
- PKC (protein kinase C), signal transduction role, 4:89
- PKC $_\epsilon$ mice, 1:5
- PKU. *See* Phenylketonuria
- Placenta
- chromosomal mosaicism, 3:79
 - stem cells, 4:30
- Plague, as bioterrorism tool, 1:69–70
- Plant genetic engineers, 3:149–150
- Plant molecular farming, 4:134–135
- Plaque, and cardiovascular disease, 1:102
- Plasma membranes
- defined, 1:113, 3:202, 4:89
 - of eubacteria, 2:12
 - structure and function, 1:105, 1:109, 1:113, 3:202
- Plasmids, 3:150–153
- cleaved by restriction enzymes, 4:33
 - Col, 3:151
 - conjugative (fertility factors), 1:140–142, 1:182–185, 1:184, 2:117, 3:151
 - defined, 1:27, 1:222, 3:16, 4:128
 - degradative and catabolic, 3:151
 - for DNA vaccines, 1:254–255
 - electrophoresis to separate, 2:49
 - as gene therapy vectors, 2:78
 - inheritance of, 2:198, 3:151
 - mobilization, 3:152
 - promiscuous, 1:183, 1:185
 - in recombinant organisms, 1:71, 2:15, 3:152
 - relaxed, stringent, and incompatible, 3:151
 - replication of, 1:152–153, 2:15, 3:151–152
 - resistance (R), 1:27, 1:140, 1:185, 2:117, 2:198, 3:16–17, 3:151, 4:69
 - structure and function, 1:140–142, 1:144, 2:14–15, 2:116, 3:150–151
 - Ti, 2:107
 - of transgenic bioremediators, 1:61
 - as vectors, 1:222, 2:107, 3:91, 3:152
 - virulence, 3:151

- Plasmin, role in blood clotting, 1:72
- Plasminogen activating inhibitor 1 (PAI-1), 1:8
- Plasminogen activators
- cost and access concerns, 1:69
 - from genetically engineered bacteria, 1:63
 - tissue plasminogen activator (tPA), 1:72
- Plasmodium falciparum*, and malaria, 2:148
- Plastics, industrial, genetically engineered, 4:135
- Plastids, defined, 3:149
- Platelets, cancerous tumors, 1:94
- Pleiotropy, 3:153–154
- albinism, 2:128, 2:206
 - cystic fibrosis, 2:206
 - defined, 2:105
 - as eugenics rebuttal, 2:20
 - impact on hormone regulation, 3:154
 - myotonic muscular dystrophy, 3:85
 - sickle-cell disease, 3:153
 - signaling pathway defects, 3:153–154
- Ploidy
- defined, 2:115
 - haplodiploid, 2:113
 - monosomy, 1:121, 1:123, 3:109
 - polyploid, 2:28, 2:69, 2:113–114, 3:97, 3:163–167
 - tetraploid, 3:97, 3:165, 3:166
 - triploid, 3:164–166, 3:164
 - See also Diploid; Haploid; Trisomy
- Pluripotency, of embryonic stem cells, 2:3
- PML bodies (PODs), 3:120, 3:124–125
- Pneumococcus* bacteria, 2:167
- transforming factors, 1:250, 1:250
- Pneumonia
- antibiotic resistance, 1:27
 - causal organism, 2:167
 - pneumocystis, 2:155
 - transformation studies, 4:122, 4:123–124
- Point estimations, 4:96
- Point mutations
- base substitutions, 3:95
 - defined, 2:26–27
 - frameshift mutations, 2:86, 3:95
 - gene evolution role, 2:26–30, 2:27
 - growth hormone gene, 2:129
 - mitochondrial diseases, 3:54
 - and proto-oncogene activation, 1:99–100, 3:128–129
- Pol* gene, 4:35, 4:36, 4:144, 4:145
- Pol I, II, and III enzymes, 4:55
- POLA* (polymerase alpha) gene, 1:231
- Polar bodies, defined, 3:29
- Polar, defined, 3:200
- Polarity, of eukaryotes. See Axis development
- Polarity, of viruses, 4:165, 4:166
- POLB* (polymerase beta) gene, 1:232
- POLD1* (polymerase delta 1) gene, 1:231–232
- POLE1* (polymerase epsilon 1) gene, 1:231–232
- POLG* (polymerase gamma) gene, 1:232
- Poliomyelitis, causal organism, 4:165
- Poliovirus
- characteristics, 4:165
 - infection outcomes, 4:169
 - replication, 4:166, 4:168
 - vaccine to combat, 4:170–171
- Pollen
- chloroplast genes lacking in, 2:108
 - defined, 1:34
 - and Mendelian ratios, 1:130
- Pollination
- artificial, 1:147
 - open, 2:146
 - outcrossing, 1:35, 1:147, 3:9
 - selfing, 1:147, 2:189, 3:9, 3:33–34
- Polly the sheep, 1:163–164
- Polyacrylamide gel electrophoresis (PAGE), 2:46–47
- 2-D, 2:48–49, 2:48, 3:207–208
 - SDS, 2:47–48
- Polyadenylated repeats, 4:9, 4:145
- Polyadenylation (poly A) addition, function, 4:59
- Polyadenylation signal, 2:53
- Polyamines, in bacteria, 1:141
- Poly-A-polymerase, 2:53
- Polychlorinated biphenyls (PCBs), bioremediation to remove, 1:59
- Polycistronic mRNAs, 3:135
- Polycyclic aromatic hydrocarbons (PAHs), 1:59, 1:99
- Polygalacturonase (*PG*) gene, 1:29
- Polygenic traits. See Complex traits
- Polyglutamine disorders (CAG_n repeats), 4:150, 4:151–152
- Polylinkers, 3:152
- Polymerase chain reaction (PCR), 3:154–159
- Archaea research tools, 1:38
- in conservation genetics, 1:187, 1:190–191
 - contamination concerns, 3:158
 - defined, 1:177, 1:235
 - degenerate primers, 3:158–159
 - in DNA profiling, 1:73
 - DNA purification tools, 3:220–221
 - in gene cloning, 1:153
 - as genotyping tool, 3:161
 - in HPLC analysis, 2:166
 - Human Genome Project role, 2:172
 - as molecular anthropology tool, 3:68
 - primers, designing, 3:155–156
 - procedures, 3:154–155, 3:155, 3:156–158, 3:156
 - reverse-transcription, 3:159
 - in SELEX analysis, 1:177
 - in STR analysis, 1:235–236
 - to study chimeras, 2:5
- Polymerases
- defined, 1:19, 3:221, 4:104
 - See also DNA polymerases; RNA polymerases
- Polymerization, defined, 1:230, 3:43, 4:109
- Polymers
- defined, 2:11, 3:112, 4:107
 - See also Transcription; Translation
- Polymorphic, defined, 1:188, 2:59
- Polymorphic markers, 2:58, 2:59, 2:60, 3:6, 3:7
- Polymorphisms, defined, 2:60
- Polymorphisms, DNA, 3:159–163
- as agricultural biotechnology tool, 3:162
 - Alu sequences, 4:145–146, 4:146
 - balanced, 1:45–46
 - banding techniques to identify, 1:135
 - cause of, 3:160
 - defined, 1:235, 2:52, 3:1, 4:11
 - as disease study tool, 3:162
 - distinguished from multiple alleles, 3:82
 - distinguished from mutations, 2:126
 - in DNA profiling, 1:234–235, 3:163
 - drug development role, 2:124
 - gel electrophoresis tools, 3:160–161, 3:161
 - genotyping, 3:160–161
 - HPLC to study, 2:166–167, 2:166

- origin of term, 3:159
 PCR tools, 3:161
 public health role, 3:216
 RFLPs, 1:236, 3:6, 3:13–14, 3:161–162, 3:161, 4:34
 selectively neutral, 3:160
 STRPs, 3:6, 3:14
 as taxonomy tool, 3:163
 VNTRs, 1:236, 3:6, 3:14, 3:162, 4:8–9
 Y chromosomes, 4:178
See also Alleles, multiple; Short tandem repeats (microsatellites)
- Polymorphisms, single-nucleotide (SNPs)**
 disease detection role, 2:118–119, 2:122, 2:166–167
 distinguished from mutations, 2:118
 in DNA comparison studies, 3:65
 in human genome, 3:162
 individual genetic variation and, 2:191
 in linkage analyses, 3:6, 3:14
 as pharmacogenomics tool, 3:146–147
- Polypeptide backbone, post-translational modifications**, 3:179, 3:179, 3:181
- Polypeptides**
 defined, 1:13, 2:84, 3:105, 4:100
 folding by chaperones, 1:116–118
 heavy and light, in antibodies, 2:179, 2:179, 3:202
 one gene-one polypeptide model, 3:103
 phytochelators, 1:60
 Rh, 1:86
 synthesis, 1:13
- Polyploidy**, 3:163–167, 3:164
 allopolyploid, 3:165–166
 in amphibians, 3:165
 autopolyploid, 3:165–166
 causes, 3:163–164
 chromosome numbers, range, 2:113–114
 defined, 2:69, 4:93
 gene evolution role, 2:28, 2:69
 genetic analysis of, 3:166
 in humans, 3:164–165
 mutations, consequences, 3:97
 in plants, 3:165–166, 3:165
 and sex determination, 3:164
 sympatric speciation, 4:93
See also Aneuploidy
- Polyspermy, consequences**, 2:35, 3:163–164
- Polytene chromosomes**, 1:126, 1:136–137, 1:136, 2:43–44
- Poplar trees, transgenic**, 1:61, 4:132
- Population**
 of Bombay, 3:173
 defined, 2:133, 3:171
- Population bottlenecks**, 3:167–171
 ancient population sizes, 3:167–168
 cheetahs, 1:189
 colonization and climatic, 3:169
 consequences, 1:188–189
 defined, 1:188
 and expansions, human evolution, 3:168–169
 genetic drift, 2:95–96
 Hardy-Weinberg equilibrium, 2:136
 right whales, 1:189
 technological and social influences, 3:169–170
See also Founder effect
- Population genetics**, 3:171–174
 balanced polymorphism, 1:45–46
 conservation biology, 1:186–190
 disease prevalence studies, 1:202–203, 1:213–215, 3:194–195
 extranuclear inheritance role, 2:194
 of eye color, 2:32
 gene flow, 2:70–71
 gene pools, 3:171–172
 genetic drift, 2:36–37, 2:70–71, 2:94–96, 3:63, 3:174
 genetic structure, 3:172
 to identify disease genes, 2:168
 inbred population studies, 2:190–191
 mutation role, 3:93
 population-based genomics companies, 2:124
 quantitative traits, 4:1–3
 speciation, 4:91–93
 statistics tools, 4:95–98
 twin registries, 4:159–161
 Y chromosomes as tools, 4:178–179
See also Founder effect; Hardy-Weinberg equilibrium
- Population parameters**, 4:96
- Population screening**, 3:175–178
 for bipolar disorder, 3:215
 for carriers, 3:176, 3:177
 criteria for, 3:175
 for cystic fibrosis, 1:202–203
vs. diagnostic tests, 3:175
 epidemiology, 2:6–7
- ethical issues, 3:177–180
 for hemophilia, 2:143
 for hypothyroidism, 3:176–177
 to identify carriers, 2:99–100
 for inherited disorders, 3:174, 3:175–176
 public health role, 3:217–219
 for sickle-cell disease, 3:177, 3:219
See also Newborn screening
- Population studies**
 defined, 2:168
See also Family studies; Population genetics
- Porin proteins**, 1:27, 1:28
- Positional cloning**
 cystic fibrosis gene, 1:202
 defined, 1:202
 as gene discovery tool, 2:59–60
 procedure, 1:155–157
- Positive selection marker systems**, 3:16–17, 4:69
- Pospiviroidae family**, 4:163, 4:163
- Posterior**, defined, 2:32
- Post-translational control**, 2:62, 3:178–182, 3:203–204
 amino acid alterations, 3:178–179, 3:180, 3:205–206
 analysis of protein structures, 3:71
 detection tools, 3:20
 exteins, 3:179, 3:181
 inteins, 3:179, 3:180
 molecular chaperones and, 1:116–118
 polypeptide backbone alterations, 3:179, 3:179, 3:181
 RNA interference, 1:31, 4:52–54
- Potassium channel genes**, 1:12, 1:13–14
- Potassium ions**, in ribozymes, 4:45
- Potato spindle tuber viroid**, 4:163
- Potatoes, genetically engineered**, 2:107, 2:108, 4:130–131
- Potts, Sir Percival**, 1:98
- Power of a study**, 4:97
- Poxviruses (*Poxviridae*)**
 characteristics, 2:117, 4:164
 replication, 4:166, 4:167
- PPD skin test, for tuberculosis**, 3:175
- Prader-Willi syndrome**, 1:123, 2:185, 2:205
- PRb (retinoblastoma protein)**, phosphorylation, 1:106–107, 1:106
- Prebiotic**, defined, 2:23
- Precursors**, defined, 1:63, 2:76, 4:135

- Predisposition (susceptibility) to disease. *See* Genetic predisposition
- Prednisone, for muscular dystrophies, 3:87
- Pregnancy
 age, and chromosomal abnormalities, 1:258, 2:97
 and breast cancer risk, 1:89–90
 ectopic and heterotopic, 4:22
 fertilization role, 2:33–36
 forced sterilization, 2:16–20, 2:90, 4:69
 gestational diabetes, 1:209, 1:212
 miscarriage, and prenatal diagnoses, 3:182, 3:184
 retinoic acid and, 1:206
 spontaneous abortions, 3:165, 4:22
 surrogacy, 1:162, 1:163–164, 4:24, 4:29
 terminating, 1:162, 2:103, 2:177, 3:187
 in utero gene therapy, 2:81, 2:82
See also In vitro fertilization; Infertility; Prenatal diagnosis; Reproductive technology
- Preimplantation genetic diagnosis (PGD)
 ethical issues, 3:186–187, 4:29–30
 uses, 2:98, 3:184, 3:185–186
- Premutations, fragile X syndrome and, 2:40–42
- Prenatal diagnosis, 3:182–187
 alpha fetoprotein test, 3:185
 amniocentesis, 1:258, 2:97, 3:183, 3:184
 chorionic villus sampling, 1:258, 2:97, 3:184
 chromosomal disorders, 1:79, 2:97
 chromosome painting, 3:183
 clinical geneticist role, 1:149–151
 cytogenetics analysis, 1:119, 3:182–185
 ethical issues, 2:81, 2:82–83, 2:103, 2:177, 3:186–187
 fetal cell sorting, 3:184–185
 genetic counseling, 2:97, 3:185, 3:186
 maternal serum marker screening, 3:185
 percutaneous umbilical blood sampling, 2:97–98
 preimplantation genetic diagnosis, 3:182, 3:184, 3:185–186, 4:29–30
 in situ hybridization tools, 3:182
 specialists, 1:74
 ultrasound scans, 3:182, 3:183, 3:185
- Prenatal diagnosis, specific diseases
 achondroplasia, 1:76
 AIS, 1:21, 1:25
 Alzheimer's disease, 1:18, 3:186
 cystic fibrosis, 1:203, 3:186
 Down syndrome, 1:257–258, 3:185
 dwarfism, 3:186
 Fanconi anemia, 3:186, 4:30
 hemophilia, 3:186
 metabolic diseases, 3:183–184, 3:186
 muscular dystrophies, 3:85, 3:86, 3:184, 3:186
 neural tube defects, 3:184, 3:185
 phenylketonuria, 2:55–56, 2:119
 Rh antibodies, 1:85
 Tay-Sachs disease, 3:184, 4:98, 4:100, 4:102
 X-linked disorders, 3:186, 3:187
- Pre-replicative complexes (pre-RCs), 4:18
- Presenilin 1 and 2 (PS1 and PS2) protein. *See* Amyloid precursor protein
- Prevalence, defined, 1:214
- Primary transcript, of RNA, 4:56, 4:57–59, 4:58
- Primates
 Alu sequences, 4:145–146
 chromosome number range, 2:113
 DNA, compared to human, 3:63, 3:64–65, 4:11
 genetic diversity, 3:167–168, 3:168
 LI elements, 4:145–146
 as organ donors, 4:142
 poliovirus in, 4:166
- Primer nucleotides, 3:155
 annealing, in cycle sequencing, 1:199
 defined, 1:44
 degenerate, 3:158–159
 in PCRs, 3:155–157
 in STR analysis, 1:236
- Primordial soup, defined, 2:23
- Printed microarrays, 1:226
- PRINTS database, 3:209
- Prion protein (PrP), conformation conversions, 3:188–189, 3:189, 3:190, 3:190
- Prions, 3:187–190
 bovine spongiform encephalopathy, 2:106, 3:188, 3:189
 Creutzfeldt-Jakob disease, 3:188, 3:189–190
 fatal familial insomnia, 3:190
- Gertsman-Sträussler-Scheinker syndrome, 3:189–190
 kuru, 3:189
 origin of term, 3:188
 protein conformation conversions, 3:188–189, 3:189, 3:190, 3:190
 scrapie, 3:188, 3:189
 structure, 3:188
 transmissible spongiform encephalopathies, 3:188–190
- Privacy, 3:190–193
 Directive on Data Protection, 3:192
 DNA samples, justice system, 3:192–193
 of gamete donors, 4:28
 genetic testing and, 1:68, 3:190
 Health Insurance Portability and Accountability Act, 2:94
 Human Genome Project concerns, 2:177
 of medical records, 3:191–192
 of pedigrees, 3:142
 pharmacogenomics and, 3:146
 of transgenic organisms, 4:130
 of twin registries, 4:160
See also Genetic discrimination
- Probability, 3:193–196
 BLAST, 2:156, 2:157, 2:212, 3:196
 Chi-square test, 3:194–195
 complex traits, 1:177–179
 and DNA profiling, 1:236, 1:238
 genetic determinism, 2:102
 genomic screening tools, 2:170
 inbreeding coefficients, 2:190
 in linkage analyses, 3:6–8
 in Mendelian genetics, 1:146–149, 3:31, 3:34–35
 p values, 3:194–196, 4:97
 Punnett squares, 1:148, 3:33, 3:35, 3:193–194
 relative risk ratios, 1:214, 2:56
See also Hardy-Weinberg equilibrium
- Probands
 defined, 3:140
 symbol for, 3:139
- Probes (molecular)
 defined, 1:88
See also *In situ* hybridization
- Procarcinogens, defined, 1:99
- Processed pseudogenes (retro-pseudogenes), 3:211–212
- Procreation
 defined, 2:17
 eugenics and, 2:17–20

- Professions**
 attorney, 1:42–43, 3:138
 bioinformatics specialist, 1:54–56
 biotechnology entrepreneur, 1:65–66, 1:72
 cellular biologist, 3:70
 clinical geneticist, 1:74, 1:149–152, 1:213–215, 2:90, 2:91–92, 3:193–194
 computational biologist, 1:54–56, 1:181–182, 3:196
 conservation geneticist, 1:190–192
 cytochemist, 2:187–189
 cytologist, 3:21
 educator, 2:1–3
 epidemiologist, 2:6–7, 2:56–57
 evolutionary biologist, 3:70
 genetic counselor, 2:89–90, 2:91–92, 3:186
 geneticist, 2:110–111, 2:212
 information systems manager, 2:192–194
 laboratory technician, 3:1–3
 microbiologist, 3:50–51
 molecular biologist, 3:70–72
 pharmaceutical scientist, 3:142–144, 3:145
 phylogeneticist, 4:41
 physician scientist, 3:147–148
 plant genetic engineer, 3:149–150
 science writer, 4:65–66
 statistical geneticist, 3:193, 3:194–195, 4:93–95
 taxonomist, 4:91
 technical writer, 4:102–104
- Progeny**, defined, 1:147, 3:127
- Progeria**. *See* Aging, accelerated: progeria
- Progesterone**
 gene expression role, 2:161
 ovaries and, 2:160
- Progesterone receptor**, function, 2:162
- Programmed cell death**. *See* Apoptosis
- Progressive era**, eugenics movement, 2:16
- Prokaryotes**
 defined, 2:11, 3:16, 4:12
 distinguished from eubacteria, 2:11
 distinguished from eukaryotes, 1:108, 2:11, 2:12
 origin of term, 1:139
 ribosomes, 2:16, 4:42–43
See also Cells, prokaryotic; Eubacteria
- Prokaryotes, genomes**
 average protein length, 2:52
 characteristics, 2:112, 2:121, 4:12
 circular DNA, 2:14–15, 2:116, 4:12
 gene expression, 2:53, 2:54, 2:62
 mapping of, 2:116
 operons, 1:142, 2:15, 3:105, 3:131–135
 organization, 2:116, 2:117
 overlapping genes, 3:135–136
 promoter sequences, 4:107
 repeated sequences, 2:117
 RNA polymerase, 4:55
 RNA processing, 2:53, 4:50, 4:57
 RNA synthesis, 4:50
 sizes and numbers, 2:116
 transposable genetic elements, 2:116
See also Chromosomes, prokaryotic; DNA replication, prokaryotes; Plasmids
- Prolactin**
 and breast cancer, 1:90
 and endocrine disorders, 2:129
- Proline (Pro)**
 chemical properties, 3:200
 genetic code, 2:85, 4:137
- Prometaphase**, 3:59
- Promoter clearance (escape)**, 4:109–110, 4:114
- Promoter DNA sequences**
 of antibody genes, 2:180, 2:182
 defined, 1:221, 2:15, 3:47, 4:10
 DNA methylation role, 3:47–48, 3:48, 3:49
 eubacteria role, 2:15
 function, 2:54, 4:56, 4:107–111, 4:108, 4:112
 gene expression role, 2:63–64, 2:64, 4:111
 impact on gene therapy, 2:77
 lacking in retropseudogenes, 4:10
 location of, 3:118
 modification, imprinting role, 2:184
 mutations in, 2:27, 2:27, 2:54, 3:100
 of operons, 3:132, 3:134
 overexpression, 2:67
 of pseudogenes, 2:30
 of retroviruses, 4:36, 4:37
 transcription factor binding to, 4:112–113, 4:113
 in transgenic plants, 4:133
- Promoter RNA sequences**, 2:53
- Promutagens**, 1:20
- Pronuclei**
 defined, 4:124
 in embryo transfers, 4:22, 4:124
- Prophages**, 2:116, 4:39, 4:119
- Prophase**
 meiosis, 3:25–26
 mitosis, 3:59
- Propionic acidemia**, symptoms and treatment, 3:40, 3:43
- Propionyl-CoA**, 3:43
- Pro-proteins (zymogens)**, polypeptide cleavage by, 3:179, 3:181
- Prosecution application procedure**, for patents, 3:138
- PROSITE database**, 3:209
- Prostaglandins**, as extranuclear hormones, 2:161
- Prostate gland**
 cancer, 1:93, 3:175
 tumors, and breast cancer, 1:91
- Prostatic antigen screening (PSA)**, 3:175
- Protanopia/anomaly**, 1:172
- Protease enzymes**
 apoptosis role, 1:209
 N-terminal signal sequence activation, 3:181
 prion disease role, 3:188–189, 3:190
 of retroviruses, 4:35
 ubiquitination role, 3:179
- Protease inhibitors**, to treat HIV/AIDS, 2:154
- Protein Design Laboratories**, humanized antibodies, 2:125
- Protein domains**
 functions, 1:24, 1:52–53, 2:30
 noncovalent interactions, 3:201
 and tertiary structure, 3:201, 3:203
- Protein folding (secondary structure)**
 α -helices, 3:188–189, 3:200–201, 3:203
 β -sheets, 3:200–201, 3:203
 by chaperones, 1:116–118, 3:203
 coiled-coil motif, 3:201
 and cystic fibrosis, 1:202
 exon shuffling and, 2:30–31
 helix-loop-helix motif, 3:201
 hydrogen bonding, 3:200–201
 predicted by computers, 2:177
 prions and, 3:187–190
- Protein kinase (PKA)**, signal transduction role, 4:89
- Protein kinase C (PKC)**, signal transduction role, 4:89
- Protein sequencing**, 3:196–198
 BLAST applications, 3:196
 carboxypeptidase tools, 3:198

- Protein sequencing (continued)
 databases, 2:156, 2:212, 3:18,
 3:198, 3:208–209
 defined, 3:197
 Edman degradation technique,
 3:197
 HPLC tools, 3:197–198
 as indicator of evolutionary relat-
 edness, 2:156–158, 2:157
 Internet tools, 2:212
 mass spectrometry tools,
 3:18–20, 3:198
 role of, 1:252
- Protein structure, 3:197
 C and N termini, 3:181,
 3:197–198, 3:207
 conformational changes, 3:204
 disulfide bridges, 3:200, 3:201
 domains, 1:24, 1:52–53, 2:30,
 3:201, 3:203
 peptide bonds, 3:200
 primary, 3:197, 3:200
 quaternary (multimeric), 3:202,
 3:203
 techniques for studying, 3:2
 tertiary, 3:201, 3:203
 tools to predict, 1:53, 1:55
See also Protein folding (sec-
 ondary structure)
- Protein synthesis. *See* Translation
- Proteinaceous infectious particles.
See Prions
- Proteinase K, 3:221
- Proteins, 3:198–204
 alternative splicing role, 1:12–14
 apparent mass, 3:207
 blotting to identify, 1:86–89
 and cell cycle regulation,
 1:105–107, 1:106
 cell death, 1:32–33
 defective, dominance relations,
 2:200–201
 degradation of, 1:33, 2:62,
 3:197–198
 digestion roles, 3:197
 DNA footprinting role,
 1:220–221, 1:221
 DNA repair role, 3:197
 drugs that target, 1:54,
 3:206–207
 engineering of, 1:64
 as enzymes, 2:52, 3:103
 gene families and, 2:29
 as gene regulators, 2:52
 human, cloned, 1:72
 immune system role, 3:197
 induced mutagenesis to study,
 3:89, 3:90–92
 interactions among, epistasis, 2:9,
 2:207
 interactions with drugs,
 2:149–150
 isoelectric points, 3:207
 isoforms, 1:13, 1:14, 1:14, 2:31,
 2:53, 2:63, 3:160
 mass, 3:207
 and membrane transport, 1:109
 methyl-binding, 3:48
 one gene-one protein model,
 3:103
 orthologs, 2:158
 packaging, 1:113
 proteomes, 2:212
 sarcomeres, 4:63
 signal transduction role, 1:110,
 2:52, 3:103, 3:197, 4:85–91
 structural roles, 3:103, 3:197,
 3:199, 4:135
 transport through cell mem-
 branes, 1:112–113
See also Post-translational con-
 trol; Proteomics; *specific proteins*
and enzymes
- Proteobacteria*, genome characteris-
 tics, 2:116
- Proteomes
 databases, 3:209
 defined, 2:212, 3:205
 distinguished from genomes,
 3:205
- Proteomics, 3:204, 3:205–209
 2-D electrophoresis tools, 2:49
 2-D PAGE tools, 3:207–208
 bioinformatics tools, 1:52–54,
 3:205
 challenges, 3:205–206
 defined, 1:52, 2:9, 3:124
 gel electrophoresis tools, 3:206
 as gene cloning tool, 1:157
 human nucleoli analysis, 3:124
 mass spectrometry tools,
 3:18–20, 3:206, 3:208
 yeast two-hybrid system, 2:9,
 3:209, 4:181
- Proteosomes, ubiquitination, 3:179,
 3:180
- Protista
 genome characteristics, 3:57
 inheritance of mitochondrial and
 chloroplast genes, 2:197
 intracellular symbionts of, 2:198
 ploidy, 2:115
- Protocols, defined, 1:19, 2:187
- Protonated, defined, 1:218
- Proto-oncogenes
 activation by retroviruses, 3:127
 activation without retroviruses,
 3:128–129
Bcr, 3:129
c-Abl, 3:129
 function, 4:152
 mutations, 1:99–100, 1:155
 transcription factors as, 3:128,
 3:130
- Protozoans
 African trypanosomes, 4:145
 as biopesticides, 1:57
 ciliated, 4:45, 4:104
 exceptions to universal code, 2:87
 as model organisms, 3:61
 ploidy, 2:115
- Provirus (proviral DNA)
 HIV, 2:152
 of retroviruses, 4:35–37, 4:39
- Prusiner, Stanley, 3:187–188, 3:188
- PS1 and PS2* (presenilin 1 and 2)
 genes. *See* Amyloid precursor
 protein
- PSA (prostatic antigen screening),
 for prostate cancer, 3:175
- Psd-95* gene, learning and memory,
 2:5
- Pseudoachondroplasia, 2:132
- Pseudoautosomal region, 1:196–197
- Pseudogenes, 3:209–213
 defined, 2:28
 distinguished from functional
 genes, 3:209–210
 evolution of, 2:28, 2:29–30,
 3:211–213
 globin gene family, 3:211,
 3:212–213, 3:212
 mutations, 2:29–30, 3:209–212
 nonprocessed, 3:210–211, 3:210
 paralogous, 3:210
 processed (retropseudogenes),
 3:210, 3:211–212, 4:8, 4:10
- Pseudohypertrophic muscles, 3:84
- Pseudomonas stutzeri*, competence
 development, 4:121
- Psychiatric disorders, 3:213–216
 ADHD, 1:39–42, 3:213, 3:215
 autism, 3:213
 bipolar disorder, 3:213, 3:214,
 3:215
 as complex traits, 1:213
 dementia, defined, 2:103
 epistatic interactions, 2:9
 eugenics and, 2:17–20
 genetic components, 3:213
 genetic counseling, 2:88–89,
 2:100
 obsessive-compulsive disorder,
 3:213

panic disorder, 3:213
 schizophrenia, 3:213, 3:214–215
 social phobia, 3:214
 Tourette's syndrome, 3:214, 3:215–216
 twin studies, 4:162
See also Alcoholism; Alzheimer's disease; Huntington's disease; Mental retardation
 PTC (phenylthio-carbamide), ability to taste, 2:191
 Public health, 3:216–220
 antibiotic resistance concerns, 1:26, 1:28, 1:185, 3:217
 approaches in genetics, 3:216–217
 communication and information role, 3:219
 epidemiology studies, 2:6–7, 3:217–218
 genetic testing role, 3:218
 Human Genome Project and, 3:215
 intervention role, 3:218–219
 and life span, 1:6
 population screening, 1:202–203, 2:6–7, 2:99–100, 2:143, 3:175–178
 surveillance activities, 3:217
 viral-disease control, 4:171
 Puffenberger, Erik, 2:190–191
 Puffer fish, as model organism, 3:61
 Puffing, of chromosomes, 1:126, 1:136–137, 1:136
 Pulse-field gel electrophoresis, 2:49–50, 4:33
 Punnett, Reginald, 3:33, 3:35, 3:36
 Punnett squares, 1:148, 3:33, 3:35, 3:193–194
 Pure-breeding lines, Mendel's experiments, 3:30
 Purification of DNA. *See* DNA purification
 Purines
 base stacking, 1:217
 in codons, wobble hypothesis, 2:85
 depurination, 1:240
 and DNA structure, 1:215–220, 3:44, 3:118–119
 metabolism disorders, 3:40, 3:44
 reactions with aflatoxins, 1:245
 structure, 3:115
 transition mutations, 3:95
 See also Adenine; Guanine
 Puromycin, resistance to, 2:72
 Puromycin-n-acetyl-transferase (*pac*) gene, as selectable marker, 2:72
 Pyloric stenosis, 1:77

Pyrimidines
 base stacking, 1:217
 in codons, wobble hypothesis, 2:85
 depyrimidination, 1:240
 dimers, as mutagens, 3:88–89
 and DNA structure, 1:215–220, 3:44, 3:118–119
 metabolism disorders, 3:40, 3:44
 structure, 3:115
 transition mutations, 3:95
 See also Cytosine; Thymine; Uracil
 Pyrococcus, inteins, 3:181
Pyrolobus fumarii, 1:38
 Pyruvate, 3:44
 Pyruvate dehydrogenase deficiency, 3:54

Q

Q-banding of chromosomes, 1:119, 1:127, 1:135
 QTL (quantitative trait loci), 4:1–3
 Quinacrine mustard, 1:127
 Quantitative trait loci (QTL), 4:1–3
 Quantitative (continuous) traits, 4:1–3
 bottom-up (measured) analysis, 4:3
 complex effects on phenotype, 4:2–3
 endophenotypes, 4:2, 4:2
 epistatic interactions, 4:1–3
 examples, 4:1, 4:2, 4:2, 4:3
 gene-environment interactions, 4:1–2
 genetic architecture of, 4:1–2
 in maize, 3:9
 top-down (unmeasured) analysis, 4:3
 See also Complex (polygenic) traits

Queen Victoria, as hemophilia carrier, 2:142, 2:144, 4:174
 Quiescent, defined, 1:163
 Quinacrine dihydrochloride, 1:127
 Quorum-sensing mechanisms, 4:121

R

Rabies virus (rhabdovirus)
 structure, 4:165, 4:170
 vaccine to combat, 4:171
 Rabson-Mendenhall syndrome, 1:211
 Racial and ethnic differences
 blood types, 3:172
 cancer, breast, 1:90, 1:91

cystic fibrosis, 1:202, 1:203
 diabetes, 1:210
 genetic testing and, 2:99–100
 genotype frequencies, 3:172
 hemophilia, 2:142
 Human Genome Project concerns, 2:175
 intelligence, 2:209
 interracial crosses, benefits, 2:147–148
 molecular anthropology tools, 3:66, 3:67
 in pedigrees, 3:141–142
 sickle-cell disease, 2:137
 thalassemias, 2:140
 See also Eugenics
 Radiation
 from atomic bombs, 2:171, 2:173
 defined, 1:99, 3:81
 hyperaccumulators of, 1:61
 Radiation genetics, 3:81
 Radiation, ionizing
 and cancer, 1:91, 1:92, 3:128
 and chromosomal aberrations, 1:119, 1:123, 3:99
 and DNA damage, 1:239, 1:242, 3:89
 dose-mutability relationships, 3:103
 Muller's research, 3:81, 3:88, 3:90, 3:91–92, 3:103
 Radiation, ultraviolet
 and cancer, 1:99, 1:100, 3:128
 and chromosomal aberrations, 1:119
 and DNA damage, 1:239, 1:244, 3:88–89
 RAG1 and 2 genes, 4:76
 Ragged red muscle fibers, 3:53, 3:53
 Ramsey, Paul, 4:26
 RAR (retinoic acid receptor), function, 2:163
 Ras and ras genes and proteins, 1:99–100, 3:128, 3:130
 Rats. *See* Rodent models
RB gene, and retinoblastomas, 3:130
 R-banding of chromosomes, 1:128
RCP (red cone pigment) genes, 1:171
In re Application of Bergy, 3:137
In re Kratz, 3:137
 Reading frames, 4:4–5
 alternate, 2:117
 disease role, 4:4–5
 open (ORFs), 1:154, 2:53, 3:209, 4:9, 4:145
 overlapping genes, 2:85–86, 3:135–136
 See also Frameshift mutations

- Receptor tyrosine kinases (RTK), 4:87–88, 4:89, 4:90
- Recessive alleles
 defined, 2:133, 3:73, 4:69
 distinguished from imprinted genes, 2:183–184
 familial adenomatous polyposis, 1:168–169
 genotype role, 2:127–128
 heterozygote advantage role, 2:147–148
 inheritance patterns, 1:147–148
 maize, 3:9
 Mendel's hypotheses, 3:31, 3:33, 3:35–36
 molecular meaning of, 2:200–201
 mutations as, 3:90, 3:93
 in polyploids, 3:166
 X-linked inheritance, 1:130–131, 3:73–74, 3:91–92
- Recessive disorders, autosomal
 albinism, 2:32, 2:128, 2:200–201, 2:202, 2:206, 2:206
 carrier testing for, 3:177
 congenital generalized lipodystrophic diabetes, 1:211
 endocrine disorders, 2:129–130
 in founder populations, 2:37
 gene therapy effectiveness, 2:74
 genetic testing to identify, 2:99–100
 Hardy-Weinberg predictions, 2:135
 hemochromatosis, 2:203
 inbreeding and, 2:189–190, 2:203
 inheritance patterns, 2:200–201, 2:202–203, 2:205
 insulin receptor gene defects, 1:211
 Laron syndrome, 2:129
 Leigh's disease, 3:54
 loss-of-function diseases, 2:200–201, 3:96
 metabolic diseases, 3:38–39, 3:45
 muscular dystrophy, limb-girdle, 3:86
 parental guilt issues, 2:88–89
 pituitary hormone deficiency, 2:129
 SCID-X1, 4:75
 selection for/against, 4:69
 stop codon mutations, 3:96
 wolfram syndrome, 1:212
 See also Cystic fibrosis; Phenylketonuria; Sickle-cell disease; Tay-Sachs disease
- Recessive disorders, X-linked, 1:213, 2:203, 4:174, 4:175
- Recipients (transplant), defined, 4:139
- Recombinant DNA. *See* DNA, recombinant
- Recombinase enzymes, 3:17
- Recombination. *See* Crossing over; Linkage and recombination (linkage analysis)
- Recombination enzymes
 transduction and, 4:119–120
 transformation and, 4:121
- Recombination fraction, 3:6–7, 3:12–13
- Recombination nodules, 1:195
- Recombining, defined, 4:174
- Red cone pigment (*RCP*) genes, 1:171
- Red-green color blindness, 1:171–172
- Reduced penetrance, defined, 2:61
- Refsum disease, 3:41
- Regulatory elongation factors, 4:111
- Rehabilitation Act of 1973, genetic discrimination and, 2:93
- Relative risk ratios
 for disease prevalence, 1:214
 gene-environment interactions, 2:56
- Relaxosomes, 3:151
- Release factors, translation, 4:138
- Renal pelvis, cancer, 1:169
- Reoviruses, replication, 4:167
- Repeated extragenic palindromic (*REP*) sequences, 2:117
- Repetitive DNA elements. *See* DNA repetitive sequences
- Replication
 cytokinesis, 3:29, 3:59
 defined, 3:46
 fission, bacterial, 3:152
 of fruit fly salivary gland cells, 2:43–44
 Hayflick Limit, 4:105
 of mitochondrial and chloroplast genomes, 2:195, 2:197, 3:56
 of plasmids, 2:198
 of prions, 3:187–190
 of retroviruses, 4:36–37, 4:37, 4:168–169
 RNA, without enzymes, 2:23–24
 rolling, 1:184
 telomere function, 4:104–105
 of viroids, 4:162–163, 4:163, 4:164
 of viruses, 1:204, 4:162
 of virusoids, 4:164
 See also DNA replication; Meiosis; Mitosis
- Replication fork, 4:13–15, 4:13, 4:14
- Replicometer mechanism, 4:105
- Replicons, 3:151
- Reporter genes. *See* Screenable markers
- Repressor (silencer) elements, 2:64, 4:111, 4:114
- Reproduction, sexual
 in Hardy-Weinberg model, 2:133–134, 2:135
 in plant propagation, 4:132
 See also Fertilization
- Reproductive technology, 4:19–26
 age concerns, 4:21, 4:23, 4:25
 and creation of twins, 4:155
 cryopreservation, 4:25
 donor insemination, 4:24
 egg donation, 4:24
 gamete intrafallopian transfer, 4:23
 intrauterine insemination, 4:24
 surrogacy, 4:24
 tubal embryo transfer, 4:23
 zygote intrafallopian transfer, 4:23
 See also Cloning organisms; In vitro fertilization; Stem cells, embryonic; Transgenic animals
- Reproductive technology, ethical issues, 4:25–26, 4:26–30
 antidiscrimination laws, 4:29
 cloning, 1:158–161
 donor insemination, 4:28
 embryonic stem cells, 2:3, 2:6, 3:186, 4:29, 4:30
 forced sterilization, 2:16, 2:90, 4:69
 gene therapy, 2:80–81, 2:82–83
 personhood of ovum, 4:27–28, 4:29
 surrogacy, 4:24, 4:29
 in vitro fertilization, 1:69, 1:201, 2:3–4, 4:28–29
 See also Eugenics
- Reptation, defined, 2:49–50
- Reptiles, sex determination, 4:81
- Resistance to antibiotics. *See* Antibiotic resistance
- Resistance to pesticides, agricultural biotechnology and, 1:11
- Resistance to pests and pathogens
 Arabidopsis research, 1:35
 breeding for, 1:149
 disease-resistant animals, 2:106
 See also Disease-resistant crops; Insect-resistant crops
- Respiratory tract disorders, birth defects, 1:75

- Restriction enzymes, 4:31–34
 as biotechnology tool, 1:70–71, 2:107, 3:114, 4:33–34
 in blotting procedures, 1:88
 defined, 1:70, 2:49, 3:152, 4:5
 as DNA fingerprinting tool, 4:32, 4:33–34
 DNA methylation role, 3:47, 4:31
 in DNA sequencing, 4:70
 in electrophoresis, 2:49
 function, 3:114, 4:31
 as gene cloning tool, 1:152, 1:154, 1:155, 1:157
 as gene mapping tool, 4:33
 nomenclature and classification, 4:31
 recombinant DNA tools, 4:5–6, 4:31, 4:33
 restriction maps, 1:155
 RFLPs detected by, 3:13, 3:161–162, 4:34
 Type I, 4:31
 Type II, 4:31–32
 Type III, 4:31
 in VNTR analysis, 1:236
- Restriction fragment length polymorphisms (RFLPs), 1:236, 3:6, 3:13–14, 3:161–162, 3:161, 4:34
- Restriction maps (genes), 1:155–156, 1:156
- Restriction point, in cell cycle, 1:104–105
- Reticular dystenesis, 4:76
- Reticuloendotheliosis virus strain T, 4:35
- Retinas
 cherry-red spot, 4:98, 4:100
 defined, 4:98
 function, 1:170
- Retinoblastoma gene (*Rb*), mutations, 1:96–97, 1:96, 3:130, 4:153
- Retinoblastoma protein (pRb), phosphorylation, 1:106–107, 1:106
- Retinoblastomas
 chromosome deletion role, 4:155
 genetic testing, 2:100–101
 mutation rates, 3:99
 PML body mutations, 3:124
 two-hit theory, 3:131, 4:153
- Retinoic acid
 concentration, 2:164
 influence on gene expression, 2:161, 2:162, 2:163, 2:164
 risk during pregnancy, 1:206
- 9-cis-retinoic acid receptor (RXR)
 function, 2:163
 nuclear receptor partners for, 2:160
- Retinoic acid receptor (RAR), function, 2:163
- Retroposition, gene evolution role, 2:28, 2:28, 2:30
- Retroposons, 3:211–212
- Retropseudogenes (processed pseudogenes), 3:210, 3:211–212, 4:8, 4:10
- Retrotransposons
 defined, 4:143
 ingi, 4:145
 LINEs, 4:8, 4:10, 4:11, 4:144, 4:145, 4:147–148
 LTRs, 4:8, 4:9–10, 4:11, 4:144, 4:145
 non-LTRs, 4:8, 4:9–10, 4:11, 4:144, 4:145
 SINEs, 4:8, 4:10, 4:11, 4:144, 4:145, 4:147–148
- Retroviridae*. *See* Retroviruses
- Retroviruses (RNA viruses), 4:34–39
 acutely transforming, 4:170
 antiretroviral therapy, 4:41, 4:41
 cancer and, 4:38–39, 4:52, 4:169
 characteristics, 2:117, 4:34–35, 4:168
 defined, 2:151
 discovery, 4:39
 endogenous, 4:35, 4:41
 ex vivo gene transfer tools, 2:76
 gag gene, 4:35, 4:36, 4:144, 4:145
 HIV as, 2:151, 4:34, 4:40–41, 4:52
 list of, 4:35
 pol gene, 4:35, 4:36, 4:144, 4:145
 proto-oncogene activation, 3:127
 provirus (proviral DNA), 4:35–37
 retrovirus-like elements, 4:41
 reverse transcriptase, 4:35–37, 4:39–42
 Rous sarcoma virus, 1:99–100, 4:39, 4:169
 simian sarcoma (SSV), 3:130
 Ty elements, 4:145
 as vectors, 2:76, 4:182
- Retroviruses, infection process
 assembly, 4:37–38, 4:37
 attachment, 4:36, 4:37
 cytopathic infections, 4:38
 egress (release), 4:37, 4:38
 infection outcomes, 4:169
 latent infections, 4:38, 4:169
 maturation, 4:37, 4:38
 penetration, 4:36, 4:37
 slowly transforming, 4:170
 uncoating, 4:36, 4:37
- Retrovirus-like elements, 4:41
- Rett syndrome
 clinical features, 2:202
- MeCP2 protein, 3:48, 3:49, 3:78
- Rev proteins, HIV-produced, 2:153
- Reverse transcriptase. *See* Transcriptase, reverse
- Reverse transcriptase inhibitors, 4:41, 4:41
- Reversed-phase columns, 2:166–167, 2:166
- RFLPs. *See* Restriction fragment length polymorphisms
- Rh blood group
 discovery, 1:84
 genetics of, 1:85–86
 as identification, 1:233
 incompatibility, 1:75, 1:84–86
- Rhabdovirus (rabies virus), structure, 4:165, 4:170, 4:171
- Rhinovirus, characteristics, 4:165
- Rhodopsin
 RHO genes, 1:171
 signal transduction role, 4:89
- Ribonuclease enzymes
 defined, 4:16
 function, 3:112, 3:114–115, 4:51, 4:57–58
 RNase H, 4:16
 tertiary structure, 3:201
- Ribonucleic acid. *See* RNA
- Ribonucleoprotein (RNA-protein) complexes, 4:42
- Ribose sugars
 discovery, 1:249
 evolution of, 2:23
 in RNA, 4:47, 4:49
 structure and function, 3:115
- Ribosomes, 1:105, 4:42–44, 4:43
 defined, 1:29, 2:13, 4:46
 eukaryotic, 4:42–43
 function, 1:112, 2:62, 3:70–71, 4:43–44, 4:50, 4:136, 4:137
 impact of antibiotics on, 1:28, 4:44
 in mitochondria and chloroplasts, 4:42
 in nucleus, 3:119, 3:125
 prokaryotic, 2:16, 4:42–43
 structure, 4:42, 4:51, 4:136, 4:137–138
 synthesis, 4:42–43
 in vivo assembly pathway, 4:43
 See also RNA, ribosomal; Translation
- Ribozymes, 4:44–46
 defined, 2:79, 4:163
 discovery, 4:44
 evolutionary origin, 4:46
 as gene therapy tool, 2:79, 4:46
 RNase P, 4:45, 4:46

- Ribozymes (continued)
 structure and function, 2:25, 4:44–45
 viroid cleavage role, 4:163, 4:163
 virusoid cleavage role, 4:164
- Ribulose 1,5-biphosphate carboxylase, 3:199
- Rice (*Oryza sativa*)
 genetically engineered, 1:10, 1:10, 2:106–108, 3:149, 4:135
 genome characteristics, 1:34, 2:121, 2:172
- Rice dwarf virus, 4:165
- Rice, George, 4:84
- Ricketts, vitamin D-resistant, 4:175
- Rickettsia prowazekii*, genome characteristics, 1:142
- Risch, Neil, 1:214
- RNA (ribonucleic acid), 4:46–52
 antisense nucleotides, 1:29, 1:30
 cytoplasmic RNPs, 4:51
 defined, 1:224
 distinguished from DNA, 4:46
 as first genetic material, 2:22–25
 oligonucleotides, 1:176–177
 polymer formation, 3:117–118, 3:118, 4:108, 4:109–110
 preserving, *in situ* hybridization, 2:188
 primary transcript, 4:56, 4:57–59, 4:58, 4:107
 of retroviruses, 2:50, 2:76, 2:112, 2:151, 4:35, 4:39–40, 4:52
 ribozymes, 2:25, 4:45–46
 in viroid genomes, 4:162–163
See also RNA processing; Transcription; Translation
- RNA, 7SL, 3:211–212
- RNA binding sites. *See* Promoter RNA sequences
- RNA catalysts. *See* Ribozymes
- RNA, complementary (cRNA), *in situ* hybridization role, 2:187–188
- RNA, double-stranded (dsRNA), 1:31, 4:53–54
- RNA editing, 4:58
- RNA interference, 1:31, 4:52–54
 dicing dsRNA, 4:53
 interference process, 4:53–54
 research applications, 4:54
- RNA, messenger (mRNA)
 alternative splicing, 1:11–13, 2:62, 2:63
 ambisense segments, 4:166
 antisense nucleotides, 1:29, 1:30, 1:97, 2:106
 of cloned organisms, 1:164–165
 coding for, 3:105
 correlation to protein levels, 3:206
 Crick's contributions, 1:193, 1:252
 defined, 1:12, 1:97, 1:206, 2:164, 4:10
 determining location of, 2:187
 function, 2:53, 2:62, 3:70, 3:114, 4:50–51, 4:57, 4:135, 4:136
 genomics studies of, 2:120
 impaired, and metabolic disease, 3:39
 instability, 2:84
 introns lacking in, 1:154, 1:157, 1:224, 2:53, 3:124, 3:211, 4:51–52, 4:110
 maturation and degradation, 2:62, 3:114–115, 4:51, 4:53, 4:107
 molecular hybridization mapping of, 1:225–226
 polycistronic, 3:135
 positive and negative polarity, 4:165, 4:166
 processed pseudogenes and, 3:211
 processing, 4:57–59, 4:58
 RNA interference, 1:31, 4:52–54
 splicing of, 4:59
 synthesis, 1:112
 synthesis of, 4:55
 translation role, 2:84, 4:44, 4:50–51
 transport from nucleus, 2:62
 of viruses, 4:165–168
- RNA polymerases, 4:54–56
 clamps, 4:56
 defined, 2:63, 2:163, 2:184, 3:134
 DNA primase, 4:13, 4:16, 4:18
 DNA-dependent, 4:168
 function, 1:112, 1:205, 2:30, 2:52, 2:54, 2:62, 4:48, 4:55, 4:56, 4:107, 4:108, 4:110
 inhibition, imprinting, 2:184
 laboratory analysis of, 4:55
 operon role, 3:133, 3:134–135
 pol I, 4:55, 4:112
 pol II, 4:55, 4:112–113, 4:145
 pol III, 4:55, 4:112
 prokaryotes, 4:55
 RNA-dependent, 4:168
 structure, 4:55–56, 4:55, 4:56
- RNA processing, 4:57–60
 in antibody genes, 2:180
 editing, 4:58
 by eukaryotes, 2:53, 4:50, 4:57, 4:58
 evolution of, 2:22–25
 exo- and endo-ribonuclease role, 4:57–58
 function, 4:49, 4:110
 gene expression role, 2:62
 of messenger RNA, 4:57–59, 4:58
 polyadenylation, 4:59
 by prokaryotes, 2:53, 4:50, 4:57
 retroposition, 2:28, 2:28, 2:30
 of ribosomal RNA, 2:53, 3:119–120, 4:57, 4:58, 4:58
 ribozyme role, 2:25, 2:79, 4:44–45
 splicing, 3:181, 4:57, 4:58, 4:59
 of transfer RNA, 4:57, 4:58, 4:58, 4:59
See also RNA polymerases; Splicing, alternative
- RNA purification, electrophoresis techniques, 2:49–50
- RNA, ribosomal (rRNA)
 banding techniques, 1:128
 of ciliated protozoa, 4:45
 coding for, 2:52, 3:105
 in eubacteria, 2:16
 function, 1:112, 4:51, 4:57
 gene clusters (tandem arrays), 1:132, 2:28, 2:68, 2:69, 2:117
 human, amount, 3:123
 in mitochondria, 3:53, 3:55–56
 peptide bond role, 4:42
 phylogenetic relationships, 1:36
 processing, 2:53, 3:119–120, 4:57, 4:58, 4:58
 as pseudogenes, 3:211
 and ribosome synthesis, 4:42–43
 as ribozyme, 4:45
 structure, 4:47–48, 4:50
 synthesis of, 3:119, 4:55
 transcription of, 3:123–124
- RNA, satellite (satellite viruses), 4:164
- RNA sequencing
 blotting procedures, 1:86–89, 1:152
 electrophoresis tools, 2:49–50
 by eubacteria, 2:16
in situ hybridization tools, 2:186–189
- RNA sequencing, mass spectrometry tools, 3:18–20
- RNA, small nuclear (snRNA), 3:211, 4:52, 4:57, 4:59
- RNA, small nucleolar (snoRNA), 4:51, 4:57, 4:59
- RNA, small temporal, 4:54
- RNA (ribonucleic acid), structure
 3' and 5' ends, 4:47, 4:49, 4:109, 4:135

- alphabet, 2:84, 4:46, 4:106, 4:135
 analysis techniques, 4:48
 base pairs, 4:47, 4:49, 4:106, 4:107
 directionality, 4:47, 4:49, 4:109
 hydrogen bonding, 2:24, 2:186, 4:47
 phosphodiester bonds, 4:47, 4:49, 4:109, 4:109
 primary, 3:115, 4:46–47, 4:47, 4:49, 4:54, 4:107, 4:109
 rudders, 4:56
 secondary, 4:47
 susceptibility to cleavage, 3:115
 tertiary, 4:47–48, 4:50
 RNA synthesis. *See* Transcription
 RNA, transfer (tRNA)
 anticodons, 2:84, 4:50, 4:137–138
 coding for, 2:52
 Crick's contributions, 1:193, 1:252
 function, 1:112, 2:84, 4:51, 4:57, 4:136, 4:137–138
 initiator, 4:137
 in mitochondria, 3:53, 3:55–56
 processing, 4:57, 4:58, 4:58, 4:59
 as pseudogenes, 3:211
 in retroviruses, 4:37
 RNase P precursor, 4:45, 4:46
 splicing of, 4:59
 structure, 4:47–48, 4:50, 4:50
 synthesis of, 4:55
 RNA triplets, defined, 1:61
 RNA tumor viruses. *See* Retroviruses
 RNA, viroid, 4:162–163, 4:163
 RNA viruses. *See* Retroviruses; Viroids; Virusoids
 “RNA world,” 4:46
 RNA-protein particles (RNPs), 4:49–50
 cytoplasmic, 4:51
 export, 4:51
 RNase
 contamination, *in situ* hybridization, 2:188
 as gene cloning tool, 1:153, 1:154
 H enzyme, 4:16, 4:17
 P enzyme, 4:45, 4:46, 4:51
 Robson and Aurbach, mustard gas studies, 3:90
 Rodent models, 4:60–62
 addiction studies, 1:5
 aging studies, 1:7
 behavior genetics, 1:48–49
 chimeras, 2:3–5, 2:5, 4:125
 cloned mice, 1:160, 1:164
 developmental processes, 1:205
 embryonic stem cell research, 2:3–6, 2:4
 as model organism, 3:60–61, 4:60, 4:61
 SCID studies, 4:75
 Tay-Sachs disease studies, 4:102
 testis-determining factor, 4:80–81
 transgenic mice, 2:4–5, 3:62, 4:61–62, 4:124–127, 4:125
 Rodent models, genomes
 characteristics, 2:120, 2:122, 3:61–62, 4:60
 DAT gene knock outs, 1:41
 DNA mutation libraries, 2:122, 4:60
 epistatic interactions, 2:8, 2:8
 FMR-1 gene knock outs, 2:42
 gene targeting, 2:73
 Hox gene knock outs, 1:208
 vs. human chromosomes, 3:96, 3:97
 mutation rates, 3:99
 PrP gene knock-outs, 3:188
 RNA interference, 4:53
 Rods (bacterial form), 2:14
 Rods, role in color vision, 1:171, 1:172
 Roe v. Wade, right to privacy, 2:92
 Rothmund's syndrome, progeroid aspects, 1:2
 Roundworms (*Caenorhabditis elegans*), 4:62–64, 4:63
 aging studies, 1:7, 4:64
 apoptosis studies, 1:32–33
 behavior studies, 3:61
 developmental processes, 1:204, 1:205, 1:206, 1:208, 4:63–64
 genome characteristics, 2:115, 2:121, 2:122, 2:172, 2:174, 4:62–63
 as model organism, 3:60, 3:61, 4:62–64
 nuclear receptor genes, 2:161–162
 restriction maps, 1:155–156
 RNA interference, 4:53
 sex determination, 4:81–82
 Rous, Peyton, 1:99–100, 3:127
 Rous sarcoma virus, and cancer, 1:99–100, 4:39, 4:169
 Roux, Wilhelm, 1:132
 Rowley, Janet, 3:127
 RPA binding protein, 4:14
 RTK (receptor tyrosine kinases), signal transduction role, 4:87–88, 4:89, 4:90
 Rubella (German measles) and birth defects, 1:75, 1:81
 vaccine, 4:129
 virus, characteristics, 4:165
 Rubin, Harry, 4:169
 Ruddle, Frank, 4:124
 RXR (9-cis-retinoic acid receptor), function, 2:163
- ## S
- Saccharomyces* sp.. *See* Yeasts
 Sachs, Bernard, 4:98
 S-adenosylmethionine (SAM), 1:242
 SAGE (serial analysis of gene expression) analysis, 1:228–229
 Saline conditions, Archaea adaptations, 1:37
 Salk, Jonas, 4:170
 Salmon, transgenic, 4:130
Salmonella choleraesuis, generalized transduction, 4:118
Salmonella, endotoxins produced by, 2:12
Salmonella Newport, antibiotic resistance, 1:28
Salmonella typhimurium, Ames test, 1:19–20, 1:20, 3:92
 Salt tolerance, transgenic plants, 2:106, 2:107, 3:149
 SAM (S-adenosylmethionine), 1:242
 Sample statistics, 4:96
 Sandhoff disease, 4:100, 4:102
 Sandhoff, Konrad, 4:100
 Sanfilippo syndrome, symptoms and treatment, 3:41
 Sanger, Fred, 4:64–65, 4:65
 DNA sequencing techniques, 1:43, 1:72, 1:198, 2:172, 4:70
 Sanger method. *See* Chain termination method
 Sarcolemma, muscular dystrophy and, 3:84–85
 Sarcomas
 defined, 1:93, 4:169
 Kaposi's, 2:151, 4:169
 Sarcomeres, in roundworms, 4:63
 Satellite sequences. *See* Short tandem repeats
 Schizophrenia, 2:9, 3:213–215, 4:162
 Schrödinger, Erwin, 1:192, 1:203
 SCID. *See* Severe combined immune deficiency
 Science writers, 4:65–67
 Scientist, defined, 3:147
 Scrapie, 3:188, 3:189
 defined, 3:188
 resistant transgenic animals, 2:106
 Screenable markers (reporter genes), 2:73, 3:16, 3:17–18, 3:91
 in situ hybridization role, 2:188

- Scurvy, cause, 1:63
- SDS (sodium dodecyl sulfate), 3:208, 3:220
- SDS-PAGE (polyacrylamide gel electrophoresis), 2:47–48
- Se* and *se* genes, 1:83–84
- Sea slugs and urchins, as model organisms, 3:60
- Secretors and nonsecretors, 1:83–84
- Segment development, homeotic genes and, 1:207–208, 2:45, 2:65–66
- Segregation analysis, defined, 4:3
- Segregation, eugenics and, 2:16
- Seipin (*GNG3*) gene, 1:211
- Seip's syndrome, progeroid aspects, 1:2
- Selectable markers
- gene cloning tools, 1:153, 3:152, 4:128
 - gene targeting tools, 2:72, 4:125–126
 - negative selection, 3:16
 - polymorphisms as, 3:162
 - positive selection, 3:16–17, 4:69
- Selection, **4:67–69**
- in antibiotics industry, 4:68–69
 - artificial, 4:67, 4:68–69
 - balanced polymorphism, **1:45–46**
 - in humans, 4:69
 - sources of variation, 4:67–68
 - See also* Eugenics; Heterozygote advantage; Hybridization (plant and animal breeding); Natural selection
- Selenium, hyperaccumulators of, 1:61
- SELEX (systematic evolution of ligands by exponential enrichment), 1:176–177
- Selfing (self-pollination)
- by *Arabidopsis thaliana*, 1:35
 - as inbreeding, 2:189
 - of maize, 3:9
 - in Mendel's experiments, 3:33–34
- Semen, abnormal, and infertility, 4:19
- Senescence, cell
- cancer and, 1:96
 - defined, 1:3, 1:96
- Senility. *See* Alzheimer's disease; Dementia
- Sequana Pharmaceuticals, gene mining by, 2:123–124
- Sequencing. *See* DNA sequencing; Protein sequencing; RNA sequencing
- Serial analysis of gene expression (SAGE) analysis, 1:228–229
- Serial endosymbiotic theory, 3:56–57
- Serine (Ser)
- chemical properties, 3:200
 - genetic code, 2:85, 4:137
 - glycosylation, 3:179
 - phosphorylation, 3:178
 - substitution, Tay-Sachs disease, 4:101
- Serine/threonine kinases, 4:87, 4:89–90
- Serotonin 1B receptor, 1:41–42
- Serotonin, as extranuclear hormone, 2:161
- Serotonin, neurotransmitter systems, 1:5
- Serum albumin, post-translational control, 3:181
- Serum cholesterol, and cardiovascular disease, 1:101
- Serum, defined, 1:82
- Severe combined immune deficiency (SCID), **4:74–78**
- ADA-SCID, 4:75, 4:77
 - bone marrow transplants, 4:75–77
 - cartilage-hair hypoplasia, 4:76
 - clinical features, 2:202, 4:75
 - David Vetter, 1:77, 4:75, 4:76
 - enzyme replacement therapy, 4:75, 4:77
 - gene therapy, 2:76, 2:81, 4:75–78
 - inheritance patterns, 4:75
 - interleukin-7 receptor (x-chain deficiency), 4:75
 - JAK3 deficiency, 4:75–76, 4:77
 - RAG1 and 2 deficiencies, 4:76
 - reticular dystenesis, 4:76
 - statistics, 4:75
 - X1 (XSCID, XL-SCID), 4:75
- Sex chromosomes, and sexual development. *See* Chromosomes, sex
- Sex determination, **4:78–82**
- AIS, **1:21–26**
 - AZF*a, *b*, and *c* genes, 4:177
 - in birds, 4:81
 - crossing-over aberrations, 1:197
 - female as default, 4:79, 4:81
 - in fruit flies, 4:81–82
 - intersex organisms, 1:25, 3:77, 3:164, 4:80, 4:82, 4:127, 4:175
 - Müllerian ducts, 1:22, 1:23, 4:79, 4:79
 - by prenatal testing, 3:187, 4:29
 - primary, 4:78
 - in reptiles, 4:81
 - in roundworms, 4:62, 4:81–82
 - secondary, 4:78–79
- SMCY* genes, 4:177
- SRY* gene and, 4:80–82, 4:126–127, 4:177
- testis-determining factor, 4:79–80
- TSPY* genes, 4:177
- Wolffian ducts, 1:21, 1:22, 1:23, 4:79, 4:79
- X chromosomes and, 2:199, 2:203, 3:76, 3:77, 3:106, 4:80–81, 4:174
- Y chromosomes and, 1:21, 1:23, 2:199, 2:203, 3:65, 3:76, 3:106, 4:80–81, 4:176–177
- Sex-linked inheritance, 2:203, 2:205
- distinguished from imprinted genes, 2:183
 - X-linked, 1:130–131, 3:73–74
 - Y-linked, 2:203, 3:65–66, 4:176–179
 - See also* X-linked disorders
- Sexual development
- AIS individuals, 1:21, 1:23–26
 - and chromosomal aberrations, 1:119, 4:80–81
 - crossing over aberrations, 1:197
 - dimorphism, 4:177
 - normal human, 1:21–23, 1:22, 3:65, 4:78–79, 4:79
- Sexual orientation, **4:83–85**
- AIS individuals, 1:23, 1:25
 - biological and genetic linkage studies, 4:84
 - defined, 1:25
 - eugenics and, 2:20
 - twin and family studies, 4:83–84
 - X chromosome role, 4:83–84
- SH2 proteins, 4:88, 4:90
- Sharp, Lester, 3:21
- Sheep. *See* Livestock
- Sherman Paradox, 2:40, 2:40
- Sherman, Stephanie, 2:40
- Shine-Dalgarno sequence, 4:138
- Shingles
- AIDS and, 2:155
 - causal organism, 4:165
- Short interspersed repeated elements (SINEs), 4:8, 4:10, 4:11, 4:144, 4:145, 4:147–148
- Short tandem repeats (microsatellites), 3:160
- cancer role, 4:11
 - defined, 1:246
 - described, 4:8
 - gene evolution role, 2:28, 2:115, 3:65
 - as molecular markers, 1:188, 3:13–14, 4:11
 - multiple alleles, 3:82–83

- as polymorphisms, 3:162
 pseudogene creation, 3:210
 satellite sequences, 4:7–9
 schematic organization, 4:10
 sequences and analysis, 1:235–236, 2:192
 in telomeres, 4:9
 VNTRs, 1:236, 3:6, 3:14, 3:162, 4:8–9
- Shull, George Harrison, 2:146, 3:10, 4:68
- Sibertia acuminata*, bioremediation by, 1:61
- Sibling recurrence ratios, 2:58
- Sickle-cell disease
 asymptomatic carriers, 2:93, 2:137, 3:177
 bone marrow transplants, 2:138
 cause, 1:239, 2:27, 2:137–138, 3:153, 3:200
 distinguished from sickle-cell trait, 2:137
 gene therapy, 2:78–79, 2:80, 2:138
 inheritance patterns, 1:213, 2:137–138, 3:142
 and malaria, 1:46, 1:203, 2:27, 2:135, 2:141, 2:148, 4:68
 medications, 2:138
 population screening, 3:177, 3:219
 as simple trait, 2:101–102
 symptoms, 2:137, 2:138, 2:202, 3:153
 as transversion mutation, 3:95
- Sight. *See* Vision
- Signal peptides, defined, 1:112, 4:85
- Signal receptors, function, 4:85–86, 4:86
- Signal transduction, **4:85–91**
 achondroplasia and, 2:130
 adenylyl cyclase, 4:88
 bioinformatics research tools, 1:56
 CAM kinases, 4:90
 cancer role, 1:94–95, 3:128, 3:129–130, 4:89–90
 CREB, 4:88, 4:89
 cyclic-AMP role, 1:5–6, 1:110, 4:89
 cytokine pathway, 4:75
 defined, 1:56, 4:86
 in eukaryotic cells, 1:110
 GPCR pathway, 4:88–89, 4:88
 GPCR-RTK pathway interactions, 4:89
 intracellular, 2:66–67
 JAK-STAT, 4:75–76, 4:77
- MAP kinase cascade, 4:90, 4:91
- N-terminal signal sequences, 3:181
- phospholipase C role, 4:89
- phosphorylation/dephosphorylation, 4:86–87, 4:87
- PKA role, 4:89
- pleiotropic effects on, 3:153–154
- protein synthesis, 2:52, 3:202–203
- RTK pathway, 4:87–88, 4:89, 4:90
- SCID role, 4:75–76, 4:77
- transcription factors and, 4:115–117
- transduction cascades, 4:86, 4:86
- vision and, 4:89
- Signals, types of, 4:85
- Silencer (repressor) elements, 2:64
- Silent mutations, 2:127, 2:157, 3:93, 3:96, 3:99
- Simian sarcoma retrovirus (SSV), and cancer, 3:130
- Simon, Theophile, 2:208
- Simple tandem repeat polymorphisms (STRPs), 3:6, 3:14
- Simple (Mendelian) traits
 and birth defects, 1:75
 cystic fibrosis, 1:177
 distinguished from complex traits, 1:177, 2:60–61
 eye color, 2:32
 genetic information about, 2:101–102
 Huntington's disease, 1:177, 1:179
See also Mendelian genetics
- Simpson, O.J., DNA profiling, 1:73
- Simpson-Golabi-Behmel syndrome, 2:132
- Sin Nombre virus, characteristics, 4:165
- SINEs (short interspersed repeated elements), 4:8, 4:10, 4:11, 4:144, 4:145, 4:147–148
- Single-gene mutation defects. *See* Simple (Mendelian) traits
- Single-nucleotide polymorphisms (SNPs). *See* Polymorphisms, single-nucleotide
- Single-strand binding proteins
 RPA, 4:14
 SSB, 4:13, 4:14, 4:18
- Sinsheimer, Robert, 2:172
- Skeletal dysplasias
 cartilage matrix protein mutations, 2:131–132
 characteristics, 2:130
 growth factor receptor mutations, 2:130
- Skin disorders
 aging and, 1:7
 cancer, 1:93, 1:99
 neurofibromatosis, 3:153, 3:154
 progeria and, 1:1, 1:3
- SLC19A2* gene, 1:212
- Sliding clamps, on DNA polymerase, 4:16, 4:17, 4:18
- Slime mold (*Dictyostelium discoideum*), as model organism, 3:60
- Sloan's achromatopsia color vision test, 1:172
- Small nuclear ribonucleoproteins (snRNP), 3:125, 4:45
- Smallpox
 as bioterrorism tool, 1:69
 causal organism, 4:165
 vaccine to combat, 4:171
- SMCY* genes, and transplant rejection, 4:177
- Smith, Hamilton, 1:71
- Smith, Michael, 3:91
- SMN (survival of motor neurons) protein, 3:124
- Smoking (tobacco)
 addiction, inheritance patterns, 1:5, 3:214
 and birth defects, 1:77
 and cancer, 1:92, 1:98, 1:100, 1:179–180
 and cardiovascular disease, 1:101
 dimethylsulfates in tobacco, 1:242
 and eye diseases, 1:179
 mutagens in smoke, 1:77, 1:92, 1:98, 1:100, 1:179–180, 1:242, 3:88–89
 and Parkinson's disease, 4:161
- SnoRNA (small nucleolar RNA), 4:51, 4:57, 4:59
- SNPs. *See* Polymorphisms, single-nucleotide
- SnRNA (small nuclear RNA), 3:211, 4:52, 4:57, 4:59
- snRNP (small nuclear ribonucleoproteins), 3:125, 4:45
- Social phobia, genetic components, 3:214
- Sodium dodecyl sulfate (SDS), 3:208, 3:220
- Soil contamination, bioremediation, 1:59–62, 1:62
- Somatic cell gene therapy, advantages/disadvantages, 2:81, 2:82–83
- Somatic cells
 diploid number, 3:24, 3:60, 3:163
 mutations, consequences, 3:93

- Somatic cells (continued)
 mutations, detecting, 3:93
 replication of, 3:58–59
 telomerase production, 4:106
 variations transmitted through, 1:104–105, 2:192
- Somatic, defined, 2:73, 3:24, 4:64
- Somatic recombination, antibodies, 2:180, 2:182
- Somatostatin, cloning, 1:72
- Soot, as mutagen, 1:98–99, 1:244
- Southern blotting
 defined, 1:86, 2:5
 procedure, 1:86–88
 to study chimeras, 2:5
 to study hemophilia, 2:143
- Southern blotting, defined, 3:221
- SOX9 gene, 4:81
- Soybeans
 as bioremediator, 1:61
 genetically engineered, 1:9, 1:10, 2:108, 4:134
- Spearman, Charles, 2:208
- Speciation, 4:91–93
 allopatric, 4:93
 biological-species concept, 4:91
 chimpanzees, 3:167–168, 3:168
 conservation genetics research tools, 1:187–189
 defined, 2:95
 ecological (sympatric), 4:93
 gene flow and, 2:70–71
 genetic drift and, 2:95–96, 4:92–93
 H. erectus distinguished from Neandertals, 3:66–67, 3:66, 3:69
 homologies as indicators, 2:158
 vs. individual genetic variation, 4:91–92
 isolating mechanisms, 4:92–93
 morphological-species concept, 4:91–92
 natural selection and, 2:156, 4:92
 repetitive DNA role, 4:10
 transposon role, 4:147–148
 See also Taxonomy (systematics)
- Speckles (interchromatin granules), 3:120, 3:125
- Sperm
 acrosome cap, 2:35, 4:21
 AZFa, *b*, and *c* genes and, 4:177
 azoospermia, 4:21
 chromosomal aberrations, 1:120
 collecting or concentrating, assisted reproduction, 4:21, 4:28
 developmental processes, 3:29
 donor insemination, 4:24
 external fertilization, 2:36
 factors inhibiting success of, 2:35
 frozen, 4:25
 fusion with egg, 2:34, 2:35
 as haploid, 2:113, 2:115, 3:24, 3:60, 3:163
 human, size, 1:108
 intrauterine insemination, 4:24
 meiosis and, 1:115, 2:33–34, 3:24–29, 3:60, 3:75
 microassisted fertilization, 4:22
 mtDNA in, 2:197
 mutation rates, 2:192
 mutations, 3:99, 4:25
 polyspermy, 2:35, 3:163–164
 SRY gene and, 3:76
 X chromosome role, 4:174
 See also Fertilization
- Sperm donation, assisted reproduction, 4:24, 4:28
- Spermatocytes, primary, 3:24
- Spider silk, in goat's milk, 1:10
- Spina bifida
 statistics, 1:75
 symptoms, 1:78
- Spinal muscular atrophy, 3:124
- Spindle apparatus
 attachment region, 1:114
 colchicine's impact on, 3:166
 defined, 1:114
 structure and function, 1:113, 1:135, 3:26, 3:59
- Spindle checkpoint errors, nondisjunction, 3:111–112
- Spinobulbar muscular atrophy (Kennedy disease), 1:26, 4:150, 4:151
- Spinocerebellar ataxia, as triplet repeat disease, 4:150
- Spinocerebellar ataxia, clinical features, 2:205
- Spirilli bacterial form, 2:13
- Spleen disorders, Gaucher disease, 3:40, 3:45, 4:4
- Splice junction mutations, Tay-Sachs disease, 4:101
- Splice sites, 1:12–13
- Splicing, alternative, 1:11–14
 amount in human genome, 3:205, 4:59
 function, 2:62, 3:106
 gene expression role, 2:62, 2:63
 genetic diversity and, 1:11–12, 2:31
 human hearing applications, 1:12
 mechanisms, 1:12–13, 1:12, 1:14
 protein isoform synthesis, 2:53
 Splicing factors, 1:13
 Splicing, ribozyme-mediated, 4:45
 Splicing, RNA, 3:181, 4:57, 4:58, 4:59, 4:101, 4:110
 Splicosomes, 1:12, 1:12, 3:124
 defined, 2:12, 4:59
 function, 4:59
 Spondylophiseal dysplasias, characteristics, 2:131
 Spongiform encephalopathies, resistant transgenic animals, 2:106
 Spontaneous, defined, 1:97
 Spontaneous mutations, 1:97, 3:90, 3:100, 4:75, 4:153
 Sporadic, defined, 1:75
 Sporophytes, defined, 2:115
 Spotfire, bioinformatics, 2:124
 Spumavirus, characteristics, 4:35
 Squamous cells, cancerous tumors, 1:94
 SRY gene, and sex determination, 1:21, 3:76, 4:80–82, 4:126–127, 4:177
 Ssa and Ssb proteins, 1:117
 SSV (simian sarcoma retrovirus), and cancer, 3:130
 Stahl, Franklin, 1:251–252
 Standard deviation, in statistical analyses, 4:96
 Stanford-Binet Intelligence Scale, 2:208
 Staphylococcus aureus, antibiotic resistance, 1:27
 STARLINK corn, 1:58
 Start codons (transcription), 2:52, 2:52
 function, 4:138
 methionine as, 2:84, 4:4, 4:137, 4:138
 overlapping genes, 3:135
 of retroviruses, 4:36
 Shine-Dalgarno sequence, 4:138
 Statistical geneticists, 3:193, 3:194–195, 4:93–95
 Statistics, 4:95–98
 association studies, 2:61
 epidemiology, 2:6–7, 2:56
 exploratory data analysis, 4:95–96
 gene-environment interactions, 2:56–57
 genomic screening tools, 2:59, 2:170
 hypothesis testing, 4:96–97
 in linkage and recombination studies, 3:7–8, 3:12–13, 3:15
 misuse, in DNA profiling, 1:238
 point estimations, 4:96
 populations and samples, 4:96

- power of study, 4:97
 quantitative trait analyses, 4:2–3
 type I errors, 4:97
 type II errors, 4:97
See also Probability
- Stem cells**
 defined, 1:103, 2:138, 4:73, 4:74, 4:106
 hematopoietic, 1:103
 as organ transplant tool, 4:141
 placental, 4:30
 replication, 1:103
 telomerase production, 4:106
 to treat immunodeficiencies, 4:74
 to treat sickle-cell disease, 2:138
- Stem cells, embryonic (ES), 2:3–6**
 chimeras, 2:3–5, 2:5, 4:125
 as disease research tools, 2:5–6
 ethical concerns, 2:3, 2:6, 3:186, 4:30
 ethical issues, 4:141
 gene knock outs/knock ins, 2:4–5
 gene targeting, 2:4, 2:73, 4:126
 human research, 2:3, 2:5–6
 mouse research, 2:3–5, 2:4
 as organ transplant tool, 4:141
 research restrictions, 1:159–160, 2:3, 4:30
- Steroid hormones**
 concentration, 2:164
 influence on gene expression, 2:159, 2:164
 regulation of, 2:161
- Steroid receptor superfamily. *See* Nuclear hormone receptor superfamily**
- Stetson and Levine, Rh system, 1:84**
- Stevens, Nettie, 3:74**
- Stigmas, defined, 1:35**
- Stomach, cancer, 1:93, 1:169**
- Stone ages, population bottlenecks, 3:169–170**
- Stoneking, Mark, 3:67**
- Stop codons (terminators)**
 factor-dependent, 4:110
 function, 2:52, 2:52, 2:53, 4:56, 4:110, 4:138
 intrinsic, 4:110
 introduction, alternative splicing, 1:13
 lacking in pseudogenes, 3:209
 mutations, consequences, 3:96
 nonsense mutations, 2:127
 nucleotides coding for, 2:85, 4:4, 4:4, 4:137
 overlapping genes, 3:135
 role of, 2:84
 universal code exemptions, 2:87
- Storage proteins, in egg cells, 4:51**
- Streisinger, George, 4:182**
- Streptomyces*, genome characteristics, 2:116**
- Streptomycin, ribosomal targets, 2:16**
- Striatum, defined, 1:41**
- Stroma, of iris, 2:32**
- Stromatolites, as oldest fossils, 2:21–22, 2:22**
- Strong, J.A., 1:119**
- STRPs (short tandem repeat polymorphisms), 3:6, 3:14**
- STRs. *See* Short tandem repeats**
- Stunting, causal organism, 4:165**
- Sturgeon, illegal trade concerns, 1:190**
- Sturtevant, Alfred Henry**
 crossing-over studies, 3:75
 fruit fly studies, 1:130–131, 2:42–43, 3:7, 3:73, 3:80
 genes as hereditary particles, 1:155
The Mechanism of Mendelian Heredity, 3:76
 New Synthesis, 3:32
- Subcutaneous, defined, 2:145**
- Subemovirus family, 4:164**
- Substance abuse. *See* Addiction; Alcoholism; Drug dependence; Smoking (tobacco)**
- Substantial equivalence principle, 4:130–131**
- Substrate deprivation therapy, Tay-Sachs disease, 4:102**
- Substrates, enzyme binding to, 3:37**
- Subterranean clover mottle virus, 4:164**
- Subtilisin enzymes, 1:63–64**
- Subzonal sperm insertion, 4:22**
- Succinylcholine, adverse reactions to, 3:144**
- Succinyl-CoA, 3:43**
- Sugars**
 defined, 1:209
 deoxyribose, 2:50, 2:51, 3:115, 3:116, 3:117
 galactose, 3:134
 glucose regulation in blood, 2:160
 glycosylation of amino acids, 3:178–179
 lactose, *lac* operons and, 2:15, 3:131–135
 in LPS layer, 2:11–12
 ribose, 2:23, 3:115
See also Deoxyribose sugars; Glucose; Ribose
- “Suicide therapy,” 2:79**
- Sulfur, as electron acceptor, 2:13**
- Sungene Technologies Laboratory, 3:150**
- Super microbes, 4:129**
- Superovulation, 4:21, 4:22**
- Superweeds, 1:11**
- Surrogate mothers**
 gene cloning, 1:162, 1:163–164
 legal and ethical issues, 4:24, 4:29
 nuclear transplantation, 1:162, 1:163–164
 transgenic mice, 4:124, 4:126
- Survival of motor neurons (SMN) protein, 3:124**
- Susceptibility genes, 1:17, 1:18**
 defined, 2:55
 genetic testing for, 2:100, 2:102–105, 2:118–120, 2:124
- Sutherland, Grant, 2:39–40**
- Sutton, Walter, 1:130, 1:149, 3:73, 3:74, 3:102**
- Swanson, Robert, 1:72**
- Sweat chloride analysis, 1:201**
- SWI/SNF gene complexes, 1:138**
- SWISS-MODEL tool, 1:53**
- SWISS-PROT database, 1:52, 3:209**
- SXR (xenobiotic receptor), function, 2:163**
- Sybase databases, 1:55**
- Symbionts**
 defined, 2:195, 3:52, 3:55
 endosymbionts, 3:52, 3:55–57
 intracellular, inheritance of, 2:198
See also Endosymbionts
- Sympatric (ecological) speciation, 4:93**
- Symptomatic genetic testing, 2:99**
- Synaptonemal complex, in crossing over, 1:194**
- Syndromes, defined, 1:74**
- Synechocystis* sp., genome characteristics, 1:142**
- Syngrafts, 4:139, 4:142**
- Syntenic genes, 3:5**
- Synthesis, defined, 3:132, 4:54**
- Syphilis, congenital, 1:75, 1:81**
- Systematic evolution of ligands by exponential enrichment (SELEX), 1:176–177**
- Systematics. *See* Taxonomy**
- Szostak, Jack, 2:25**

T

- T cell receptors, function, 2:179, 2:181, 2:181**
- T cells**
 CD4T, 2:152, 2:153, 2:154, 2:155

- T cells (continued)
 diversity of, 2:179
 function, 2:153, 2:178, 2:181, 4:38
 interactions with MHC, 2:181, 2:183, 4:139–140, 4:141
- T4 bacteriophage, structure, 4:165–166
- Tamoxifen, and breast cancer, 1:90
- Tandem repeats. *See* Short tandem repeats
- Taq DNA polymerase, 1:38, 1:199, 3:157
- Taq enzymes, 1:199
- Tat proteins, HIV-produced, 2:153
- TATA-binding protein (TBP), function, 4:112
- Tatum, Edward, 1:141, 1:185, 3:102, 3:103
- Taxonomists, defined, 4:91
- Taxonomy (systematics)
 conservation genetics applications, 1:186–189, 1:190
 defined, 1:190
 DNA and protein sequence indicators, 2:156–158
 of eubacteria, 2:16
 molecular anthropology role, 3:63–70
 molecular systematics, 2:158
 polymorphisms as tools, 3:163
 restriction enzyme tools, 4:33–34
 transposons as indicators, 4:146
See also Speciation
- Tay, Warren, 4:98
- Tay-Sachs disease, 4:98–102
 in Ashkenazic Jews, 2:37, 2:99–100, 3:174, 4:98, 4:101–102
 clinical features, 2:202, 4:98–100
 founder effect, 2:37, 3:174, 4:101
 gene therapy, 4:102
 genetic testing, 2:96, 2:99–100, 3:174, 3:177, 3:184, 4:98, 4:101–102
 heterozygote advantage, 4:101
HEXA and *HEXB* genes, 4:98, 4:99, 4:99, 4:100–101
 history, 4:98
 inheritance patterns, 2:203, 4:98
 molecular biology of, 4:100
 mouse models, 4:102
 prenatal diagnosis, 4:98, 4:100, 4:102
 symptoms and treatment, 3:40, 3:45
- TB. *See* Tuberculosis
- TBP (TATA-binding protein), function, 4:112
- T-cell leukemia, 4:38, 4:165, 4:169
- T-cytotoxic cells, 4:140
- T-DNA (DNA, transformed), 1:35
- Technical writers, 4:102–104
- Telomerase enzymes
 function, 1:135, 4:41, 4:105–106
 hTERT, 4:106
- Telomerase gene, mutations, 1:96
- Telomere shortening
 aging and, 1:3, 1:7, 1:160, 1:164, 4:105, 4:106
 cell senescence and, 1:96
 cloning implications, 1:160, 1:164
 discovery, 4:105
 and progeria, 1:3
- Telomeres, 4:104–106
 capping/uncapping proteins, 4:105
 defined, 1:7, 2:115, 3:121, 4:17
 discovery, 4:105
 function, 1:144, 4:104–105
 repetitive sequences, 4:8–9, 4:10, 4:104, 4:105
 replication of, 4:17, 4:104
 structure, 1:135, 1:144, 3:121, 4:104, 4:105
 terminal restriction fragments, 4:104
 T-loops, 4:104
- Telomeric repeat amplification protocol (TRAP), 4:106
- Telophase
 I, meiosis, 3:26
 II, meiosis, 3:27
 mitosis, 3:59
- Temin, Howard, 4:35, 4:39–40, 4:169
- Temperature extremes, Archaea adaptations to, 1:36–38
- Templates
 defined, 3:155, 4:48
 for fluorescent automated sequencers, 4:73
vs. nontemplate strands, 4:107
 for PCR reactions, 3:155–158
 in transcription, 4:48–49, 4:54, 4:107
- Tenets, defined, 3:177
- Teratogens
 alcohol, 1:81
 congenital syphilis, 1:81
 defined, 1:80
 industrial toxins, 1:77
 isotretinoin (Accutane), 1:80–81
 medications, 1:80–81
 recreational drugs, 1:81
 risk for birth defects, 1:80
 rubella, 1:81
 thalidomide, 1:80
 in tobacco, 1:77
 toxoplasmosis and, 1:81
- Terminal restriction fragments (TRF), 4:104
- Termination factors, 4:110
- Termination sites/terminators. *See* Stop codons
- Terrorist attacks, September 11, 1:70, 1:234
- Test crosses, 1:147–148
- Testes
 cancer, 1:25, 1:93, 4:177
 development, AIS and, 1:23, 1:25
 development, normal human, 1:21–23, 4:78–79, 4:79, 4:80–81
 function, 2:160, 3:24, 3:60
 Müllerian ducts, 4:79, 4:79
- Testicular cords, 4:78
- Testis-determining factor, 4:79–80
- Testosterone
 gene expression role, 2:67, 2:161–163
 as Klinefelter's syndrome treatment, 1:79
 regulation, AIS, 1:23–26
 regulation, normal development, 1:21–23
 signal transduction role, 1:110
 testes and, 2:160, 4:79
 transcription factors and, 4:115
- Tetanus (lockjaw), causal organism, 2:13
- Tetracycline
 resistance to, 1:26, 1:28, 1:185
 ribosomal targets, 2:16
- Tetrads, in crossing over, 1:194
- Tetrahymena*
 gene activation, 1:138
 as model organism, 3:61
 telomere sequence, 1:135, 4:104
- Tetraploidy, consequences, 3:97
- TFII-D (transcription factor D for RNA polymerase II), 2:63, 4:112–113, 4:113
- Thalassemias
 alpha, 2:140
 beta, 2:140
 blood transfusions to treat, 2:140
 bone marrow transplants, 2:140–141
 causes, 2:139–140, 2:148
 gene therapy, 2:140
 inheritance patterns, 2:140, 2:148
 intermedia, 2:140

- major, Cooley's anemia, 2:140
and malaria, 2:141, 2:148
transposon recombinations and, 4:11
- Thale cress. *See Arabidopsis thaliana*
- Thalidomide, and birth defects, 1:80
- Thanatophoric dysplasia, characteristics, 2:130
- T-helper cells, 4:140
- Thermodynamics, defined, 4:48
- Thermotoga maritima*, genome characteristics, 1:142
- Thermus aquaticus*, 1:38, 1:199, 3:157
- Thiamine, to treat TRMA, 1:212
- Thiamine-responsive megaloblastic anemia syndrome (TRMA), 1:212
- Thiogalactoside acetyltransferase enzyme, 3:132, 3:135
- Thioguanine, 3:93
- Threatened population status, 1:186, 1:190
- Threonine (Thr)
genetic code, 2:85, 2:87, 4:137
glycosylation, 3:179
phosphorylation, 3:178
serine/threonine kinases, 4:87, 4:89–90
- Thrush disease, 2:155
- Thymidine kinase (TK), in selectable markers, 3:17, 4:125–126
- Thymine
depyrimidination, 1:240
in DNA alphabet, 2:83, 4:106
and DNA structure, 1:215–220, 1:250–251, 2:50, 2:51, 3:94
evolution of, 2:22–24
mutagenic base analogs, 3:87
origin of term, 1:249
replaced by AZT, 2:155–156
structure, 1:216, 3:115, 3:116, 3:118, 3:119, 3:119
See also Base pairs
- Thyroid glands, function, 2:160
- Thyroid hormone
concentration, 2:164
function, 2:160
influence on gene expression, 2:161, 2:164
- Thyroid-stimulating hormone (TSH), and endocrine disorders, 2:129–130
- Ti plasmids, 2:107
- TIGR (The Institute for Genomic Research), 2:173
- Time-of-flight (TOF) tubes, 3:20
- Timofeeff-Ressovsky, Nikolay, 3:103
- Tissue culture cells, as clones, 1:161
- Tissue donors, legal ownership concerns, 3:3–4
- Tissue plasminogen activator (tPA), 1:72
- Tissue scaffolding, 4:142
- TK gene, gene targeting studies, 4:125–126
- TK (thymidine kinase), in selectable markers, 3:17
- T-lymphocytes, 4:74–76, 4:78, 4:140
- TNT (trinitrotoluene), bioremediation to reveal, 1:62
- Toba volcano, 3:168
- Tobacco
transgenic, firefly protein, 1:69, 4:130
See also Smoking
- Tobacco mosaic virus, replication, 4:168
- Togaviruses, replication, 4:168–169
- Tomato bushy stunt virus, 4:165
- Tomato plant macho viroid, 4:163
- Tomatoes
Flavr-Savr, 1:29, 1:73, 2:107
frost- and salt-tolerant, 2:107
genetically engineered, 4:6
- Tongue-rolling trait, 2:191, 2:191
- Top-down (unmeasured) genotype strategy, 4:3
- Topoisomerase enzyme
pseudogenes for, 2:30
replication role, 4:14–15, 4:18
- Topological, defined, 3:114
- Toshiba, genetic microchips, 2:119
- Tourette's syndrome
genetic components, 3:214
OCD and ADHD co-morbidity, 3:215
symptoms, 3:215
- Toxic-gain-of-function mutations, 2:201
- Toxoplasma gondii*, 1:81
- Toxoplasmosis, 1:81
- TP53 gene. *See p53* gene
- TPA (tissue plasminogen activator), 1:72
- Traits. *See* Phenotypes
- Trans regulatory elements, 4:114
- Transcriptase, reverse, 4:39–42
AIDS and, 4:40–41
defined, 3:159, 4:105
DNA library role, 1:223, 1:224, 4:6
as gene cloning tool, 1:153, 1:154, 4:40, 4:128
as gene discovery tool, 4:40
of HIV, 2:151, 2:152–153, 2:154, 2:155
how it works, 4:39–40, 4:40
inhibitors, 4:41, 4:41
of retroviruses, 4:168
telomerase enzymes, 1:135, 4:41, 4:105–106
- Transcriptase, viral, 4:168
- Transcription, 4:106–111
of antibody genes, 2:180, 2:182
assembly on DNA, 4:108
chromatin condensation/decondensation, 1:137–139, 2:63, 2:65, 2:164, 3:25
defined, 1:221, 2:10, 3:60, 4:4
determining location of, 2:187, 4:50
directionality, 3:118, 3:155–156, 4:135
DNA footprinting tools, 1:221
DNA methylation and, 3:48
E. coli as research tool, 2:10
elongation phase, 4:56, 4:108, 4:110
energy requirements, 1:111
gene activation and repression, 1:137–139, 1:225, 2:15–16, 2:64
histone acetyltransferases and, 1:137–138
hormonal regulation of, 2:159, 2:162, 2:163–164
inhibition, imprinting, 2:184
initiation phase, 4:108, 4:109–110, 4:111, 4:112
interference by DNA crosslinks, 1:245
negative regulation, 4:111
of operon genes, 3:133, 3:134–135
pause sites, 4:110
positive regulation, 4:111
procedure, 2:62–63, 2:62, 3:70, 4:108, 4:135
retroposition, 2:28, 2:28
RNA transcript role, 1:229, 4:107
of rRNA genes, 3:123–124
termination phase, 4:56, 4:108, 4:110
See also Promoter DNA sequences; RNA polymerases; Splicing, alternative; Splicing, RNA; Start codons; Stop codons
- Transcription factors, 4:112–117
defined, 1:23, 2:27, 3:48, 4:38
encoding for, 1:205
in fruit flies, 2:45
function, 2:54, 3:199, 4:56, 4:112
gene expression role, 2:63–64
gene-specific, and transcription rates, 4:113–114

- Transcription factors (continued)
 helix-turn-helix DNA-binding motif, 4:115, 4:116
 hormone receptors as, 2:161
 huntingtin's interference with, 4:152
 and morphogen gradients, 1:206
 mutations, and endocrine disorders, 2:129
 as oncogenes, 3:128, 3:130
 organism development role, 1:205, 1:207–208
 point mutations and, 2:27
 protein phosphorylation role, 1:106–107, 1:106
 regulation of, 4:115–117
 testosterone as, 1:110
 triplet repeats, 4:149
 zinc-finger DNA-binding motif, 4:114, 4:115
- Transcription factors, specific
 acidic activation proteins, 2:64, 2:66
 B, 4:112
 basal, 4:112–113, 4:113
 c-Fos, 3:130
 c-Jun, 3:130
 PIT1, 2:129
 POU2F1, 2:129
 PRO1, 2:129
 TBP, 4:112
 TFII-D, 2:63, 4:112–113, 4:113
- Transcription, reverse
 discovery of DNA, 4:39–40
 hepatitis B virus, 4:41
 in human genome, 4:41
- Transcription, specific organisms
 bacteriophages, 4:118–120
 eubacteria, 2:15–16
 eukaryotes, 2:53, 4:50
 HIV, 2:153, 2:154, 2:155–156
 mitochondria, 3:56
 viruses, 4:166–169
- Transcripts, RNA
 defined, 1:229
 function, 4:107, 4:110
- Transducers, generalized, 2:15
- Transducing particles, 4:118
- Transduction, by bacteriophages, 1:182, 2:15, 4:117–120
 DNase and, 4:123
 generalized, 4:117–118, 4:118
 genome mapping, 4:119–120
 prophages, 2:116, 4:39, 4:119
 specialized, 4:118–119
- Transduction cascades. *See* Signal transduction
- Transduction, signal. *See* Signal transduction
- Transfer (*tra*) genes, of plasmids, 3:151, 3:151
- Transfer RNA. *See* RNA, transfer (tRNA)
- Transformation, 4:121–124
 artificial, 4:121, 4:123
 bacteria-mediated, 1:9, 1:35, 1:153, 2:15, 3:152
 DNase and, 4:123
 function, 1:182, 4:121
 Griffith's experiment, 4:122, 4:123–124
 marker systems, 3:15–18
 natural, 4:121
- Transfusions
 blood group characteristics, 1:83
 and Rh antibodies, 1:84–86, 1:85
- Transgenes
 defined, 1:68, 2:4, 3:17
 in gene therapy, 2:77
 in marker systems, 3:15–18
- Transgenic animals, 4:124–127
 agricultural biotechnology, 1:10–11, 1:73
 developmental process studies, 1:204–205, 1:208
 embryonic stem cells and, 2:4–6, 4:126
 mice, 2:4–5, 3:62, 4:61–62, 4:124–127
 salmon, 4:130
 selection of targeted cells, 4:125–126
 targeted gene replacement, 4:125
 uses, 2:106, 4:124, 4:126–127, 4:130
See also Gene targeting; Knock-in mutants; Knock-out mutants
- Transgenic microorganisms, 4:127–129
 as bioremediators, 1:61–62, 4:129
 research history, 1:71
 transposons as tools, 4:148
 uses, 1:62–63, 1:66–70, 4:129
 yeasts, 1:62, 4:180–181
See also Cloning genes
- Transgenic organisms
 defined, 1:66, 4:129–130
 patent issues, 3:136–138
- Transgenic organisms, ethical issues, 4:129–132
 damage caused by activists, 4:132
 economic concerns, 4:131–132
 environmental concerns, 1:66–68, 4:129, 4:130
 food safety concerns, 1:11, 2:107, 2:108–109
 privacy concerns, 1:66, 4:130
See also Biotechnology, ethical issues
- Transgenic plants, 4:132–135
 agricultural biotechnology, 1:9–11, 1:64
 antigens in, 4:134
 antisense nucleotide tools, 1:29, 2:106
Arabidopsis research, 1:35
 biopesticides, 1:57–58
 corn, 1:9, 1:58, 1:68, 2:108, 3:149, 4:130
 direct DNA absorption, 4:133
 disease-resistant crops, 2:106, 2:107, 4:134
 distinguished from controlled breeding, 2:106
 extended shelf life, 4:133, 4:134
 genetic modification techniques, 2:107–108
 genetically modified, traits, 2:106–107
 herbicide-resistant crops, 1:58, 4:131–132, 4:134
 industrial enzymes, 4:134–135
 insect-resistant crops, 4:134
 particle-mediated transformation, 4:132–133
 patenting, 1:73
 regulatory concerns, 2:107
 tobacco, 1:69, 4:130
 uses, 1:62, 1:63
 vector-mediated transformation, 4:132
- Transgenics, defined, 4:182
- Transient neonatal diabetes (TNDM), 1:212
- Transition mutations, 3:95
- Translation, 4:135–139
 alternative splicing, 2:31, 2:53, 2:62, 2:63
 defined, 1:29, 2:10, 3:60, 4:42
 determining location of, 2:187
E. coli, 2:10
 elongation phase, 4:136, 4:138
 initiation phase, 4:136, 4:137–138
 procedure, 1:112, 1:225, 2:52–53, 2:84, 3:70–71, 4:43–44, 4:50–51
 pseudogenes, 2:30
 regulatory mechanisms, 2:62
 release phase, 4:136
 signal sequences, 3:202–203
 termination phase, 4:136, 4:138
 transplastomics techniques, 2:108

- See also Post-translational control; Proteins; Ribosomes; RNA, messenger (mRNA); RNA, ribosomal (rRNA); RNA, transfer (tRNA)
- Translation, specific organisms
bacteriophages, 4:118–120
eubacteria, 2:16
eukaryotes, 1:112
HIV, 2:153, 2:154
prokaryotes, 2:53
- Translocation
and chromosomal aberrations, 1:122–123, 1:124, 3:96–97, 3:127
defined, 1:123, 2:59, 3:81
impact on tumor suppressor genes, 4:154
and proto-oncogene activation, 3:128–129
in pseudogenes, 3:211
signal sequencing role, 3:202
See also Crossing over (recombination)
- Transmembrane conductance regulator gene, cystic fibrosis, 4:127
- Transmissible spongiform encephalopathies (TSE), 3:188–190
- Transplantation, 4:139–143
allografts, 4:139, 4:142
autografts, 4:139, 4:142
donor, defined, 4:139
donors, legal issues, 3:3–4, 4:141
ethical issues, 4:141, 4:142–143
lungs, 1:201
minor histocompatibility genes, 4:140
recipient, defined, 4:139
rejection-prevention strategies, 4:141
SMCY genes and, 4:177
syngrafts, 4:139, 4:142
T cell-MHC interactions, 2:181, 2:183, 3:82, 4:139–140, 4:141
tissue scaffolding, 4:142
T-lymphocyte response, 4:140
to treat metabolic diseases, 3:39, 3:45
xenografts, 4:139, 4:142–143, 4:142
See also Bone marrow transplants; Nuclear transplantation
- Transplastomics, 2:108
- Transposable genetic elements, 4:143–148
Agrobacterium-mediated transformation, 1:9, 1:35
antibiotic resistance, 1:27–28
and cystic fibrosis, 1:202
defined, 2:43, 3:47, 4:53
described, 4:7, 4:143
DNA methylation role, 3:47–48
DNA transposons, 4:8, 4:9–10, 4:11, 4:143
double-stranded forms, 4:53
E. coli-mediated transformation, 1:153
as gene cloning tool, 4:148
gene evolution role, 2:30, 3:97, 4:145–146
gene expression role, 3:211–212
and hemophilia, 2:31, 2:143, 3:97, 4:146
horizontal transmission, 4:145, 4:146
IS elements, 2:116, 2:117, 4:144–145, 4:144
LI, 4:145–146
LINEs, 4:8, 4:10, 4:11, 4:144, 4:145, 4:147–148
McClintock's hypothesis, 3:10, 3:21–22, 3:97, 4:143–144
MITEs, 4:9
as mutagens, 3:91, 3:97
non-replicative mobility, 4:147
recombination role, 4:148
replicative mobility, 4:147
retrotransposons, 4:143
retrotransposons, LTR, 4:8, 4:9–10, 4:11, 4:144, 4:145
retrotransposons, non-LTR, 4:8, 4:9–10, 4:11, 4:144, 4:145
schematic organization, 4:11
SINEs, 4:8, 4:10, 4:11, 4:144, 4:145, 4:147–148
speciation role, 4:147–148
as vectors, 2:43, 3:91, 3:97–98, 4:146
vertical transmission, 4:145–146, 4:145
See also Alu sequences
- Transposable genetic elements, specific organisms
bacteriophages, 4:144
fruit flies, 2:43, 3:97–98, 4:145, 4:146, 4:148
humans, 2:30, 2:67–68, 3:64, 3:97, 4:9–10, 4:145–146
mammals, 4:145–146
primates, 3:64, 4:145–146
prokaryotes, 2:116, 2:117, 4:144–145
retroviruses, 4:145
Transposase enzymes, 4:144
Transposon tagging, 4:148
Transposons. See Transposable genetic elements
- Transversion mutations, 3:95
- TRAP (telomeric repeat amplification protocol), 4:106
- Treponema pallidum*, genome characteristics, 1:142
- TRF (terminal restriction fragments), 4:104
- TRiC/CCT chaperones, 1:117
- Trichromat color vision defects, 1:171
- Trilaminar, defined, 1:114
- Trinitrotoluene (TNT), bioremediation to reveal, 1:62
- Triplet code. See Codons; Genetic code
- Triplet repeat diseases, 4:148–152
defined, 2:100, 4:4
expansion mutations, 4:149–150, 4:149
gene cloning tools, 4:152
genetic testing, 2:100, 4:152
noncoding gene disorders, 4:9, 4:149, 4:150
See also DNA repetitive sequences
- Triplet repeat diseases, specific
dentatorubral-pallidolusian atrophy, 4:150
Fragile XE syndrome, 4:150
Friedreich's ataxia, 4:150
Kennedy disease, 1:26, 4:150, 4:151
myotonic muscular dystrophy, 2:42, 3:86, 4:150
polyglutamine disorders, 4:150, 4:151–152
spinocerebellar ataxia, 2:205, 4:150
See also Fragile X syndrome; Huntington's disease
- Triplet repeat sequences, 4:148–149
expansion, slipped-strand DNA, 1:218
mutations, consequences, 3:96
- Triploidy, 3:164
in animals, 3:164–165
defined, 3:164
- Trisomy
chromosomal mosaicism, 3:79
consequences, 3:109, 3:111
defined, 1:121, 3:10, 3:79
genetic testing, 2:97, 2:98
See also Down syndrome
- Tristan da Cunha islanders, inbreeding, 2:190, 3:174
- Tritium, radioactive, to map genes, 1:139
- TRNA. See RNA, transfer

- Trypanosome brucei*, mRNA editing, 1:13
- Trypanosomes, African, 4:145
- Tryptophan (Trp)
genetic code, 2:84, 2:85, 2:87, 4:137
operons, 3:132, 3:134
- TSC1* and *TSC2* genes, and tuberous sclerosis, 4:153
- Tschermak, Erich, 3:102
- TSE (transmissible spongiform encephalopathies), 3:188–190
- TSH (thyroid-stimulating hormone), and endocrine disorders, 2:129–130
- TSPY* genes, 4:177
- Tubal embryo transfer, 4:23
- Tuberculosis (TB)
antibiotic resistance, 1:26, 1:28, 1:185
DNA vaccines, 1:255
eugenics and, 2:17
genetic components, 1:213
PPD skin test, 3:175
Tay-Sachs disease and, 4:101
- Tuberous sclerosis, genomic screening to identify, 2:168
- Tumor suppressor genes, 4:152–155
apoptosis and, 4:154
cancer and, 1:29, 1:91, 1:95, 1:99–100, 1:155
discovery, 3:130
function, 4:152, 4:154, 4:169
loss-of-function mechanisms, 4:154–155
multiplicative effects, 1:180
mutations of, 3:98, 4:152–155
as recessive, 3:130
sequential activation of, 3:131
- Tumor suppressor genes, specific
APC, 1:167–168, 4:153
BRCA1 and *BRCA2*, 4:153
CDKN1C, 4:153
MEN1, 4:153
NF1 and *NF2*, 3:130, 3:153, 4:153
p53, 1:95, 1:167–168, 2:5, 3:98, 3:130, 4:153–155, 4:169
TSC1 and *TSC2*, 4:153
- Tumors
benign *vs.* malignant, 1:92–93
defined, 1:107, 4:152
gene expression patterns, 1:227
gene mutations in, 1:94–97
immune system response to, 2:79
telomerase production, 4:106
unregulated cell growth, 3:153
Wilms, 2:132, 2:185
- See also* Cancer; Carcinomas; Oncogenes
- Turner, William, 4:84
- Turner's syndrome
cause and symptoms, 1:121, 3:111, 4:80, 4:174
hormone therapy, 1:79
karyotype, 1:122
progeroid aspects, 1:2
- Turpin, M. Raymond, 1:119
- Twin registries
Australian, 4:84
Elderly African-Americans, 4:159
privacy concerns, 4:160
twins as volunteers, 4:159
World War II Veterans, 4:159, 4:160
- Twin studies
addiction, 4:162
ADHD, 1:40
alcoholism, 1:4–5, 1:50–51, 4:96–97, 4:162
Alzheimer's disease, 1:16
behavior, 1:47–48, 1:47, 1:51, 4:157–158, 4:162
benefits, 4:160
bipolar disorder, 3:215
cardiovascular disease, 1:101, 4:162
concordance, disease, 1:214, 2:168
concordance rates, 2:58, 4:158, 4:158
intelligence, 2:207–208, 2:209–210
Parkinson's disease, 4:159, 4:160–161
psychiatric disorders, 3:213, 4:162
schizophrenia, 3:214–215, 4:162
sexual orientation, 4:83–84
standards, 4:160
Tourette's syndrome, 3:215
See also Adoption studies
- Twins, 4:155–162
conjoined (Siamese), 4:156
developmental mechanisms, 2:35–36
epistatic interactions, 2:9
identical *vs.* fraternal, 4:156–157
mitochondrial inheritance, 4:157
- Twins, fraternal
defined, 1:16
developmental mechanisms, 4:155
statistics, 4:155
- Twins, identical
and cloning, 1:158, 1:160, 1:161, 1:165
defined, 1:16
developmental mechanisms, 4:155–156, 4:156
MHC compatibility, 4:140
mosaicism, 3:79
similarities and differences, 4:157–158
statistics, 4:156
- Two-dimensional polyacrylamide gel electrophoresis (2-D PAGE), 2:48–49, 2:48, 3:207–208
- Ty elements, in retroviruses, 4:145
- Type A syndrome, 1:211
- Type I and II errors, in statistical analyses, 4:97
- Typological (essentialist) view of evolution, 4:92
- Tyrosine (Tyr)
breakdown, alkaptonuria, 3:43
genetic code, 2:85, 4:137
kinases, 4:87–89
phosphorylation, 3:178
RTK pathway, 4:87–88, 4:89, 4:90
- Tyrosine kinase, 3:129
- Tyrosinemia, symptoms and treatment, 3:40
- ## U
- Ubiquitin, 3:179
- Ubiquitination, of proteins, 3:179, 3:180
- Uid A (B-glucuronidase *gus* A) enzymes, 3:18
- Ulcers, and blood group, 1:84
- Ultrasound scans, 3:182, 3:183, 3:185, 4:21, 4:24
- Uniparental disomy, impact of, 2:185
- Uniparental inheritance, defined, 2:197
- Unmeasured (top-down) genotype strategy, 4:3
- Uracil
evolution of, 2:22–24
origin of term, 1:249
in RNA alphabet, 4:46–47, 4:106
structure, 3:115, 3:116, 3:119, 4:48, 4:49
U-G base pairs, premutagenic, 1:241–242
See also Base pairs
- Urea, formation disorders, 3:41, 3:45
- Ureters, cancer, 1:169
- Uric acid, and gout, 3:44
- U.S. Department of Agriculture (USDA)

- biotechnology regulations, 1:11, 2:107, 4:131
 Human Genome Project role, 2:173
 U.S. Patent and Trademark Office (USPTO), 1:42, 2:176
 U.S. Registry of Elderly African-American Twins, 4:159
 Uterus
 cancer, 1:93
 development, 1:22, 4:79
- V**
- Vaccines
 classic, 1:253–254
 developing, 3:71
 DNA, 1:254–255, 2:121, 4:129
 from genetically engineered plants, 4:134
 for HIV, 2:151
 for measles, 4:129
 from recombinant DNA, 4:6
 for viral diseases, 4:170–171
- Vacuoles, 1:105
 defined, 1:60
- Vagina
 development, 1:22
 function, 2:35
- Valine (Val)
 chemical properties, 3:200
 genetic code, 2:85, 4:137
 substitution, sickle-cell disease, 2:137, 2:139, 2:148, 3:200
- Van der Waal's forces
 in base-pair stacking, 1:217
 defined, 1:217
- Variable expressivity, 2:201, 2:204
- Variable number of tandem repeats (VNTRs)
 described, 3:162
 in DNA profiling, 1:236
 in linkage analyses, 3:6
 as mapping tool, 3:14
 minisatellites, 4:8–9
 schematic organization, 4:10
- Variable regions, antibodies, 2:179, 2:180
- Variegate porphyria, as autosomal dominant disorder, 2:201
- Variola virus, characteristics, 4:165
- Varmus, Harold, 4:38
- Vas deferens
 cystic fibrosis and, 1:200, 1:201
 development, 1:21, 1:22, 1:23
- Vaso-occlusive crises, 2:138
- Vasopressin, post-translational control, 3:181
- Vector-mediated transformation, 4:132
- Vectors
 artificial chromosomes as, 1:145–146
 defined, 1:70, 2:4, 3:91, 4:5
 DNA library tools, 1:222–224
 expression cloning, 1:157
 gene cloning, 1:70–71, 1:152–153, 2:4, 4:128, 4:132
 gene targeting, 2:71, 2:72, 4:125–126
 gene therapy, 2:74–77
 plasmids as, 1:222, 2:107, 3:91, 3:152
 recombinant DNA, 4:5–6
 retroviruses as, 4:182
 for site-directed mutagenesis, 3:91
 for transgenic plants, 2:107
 transposons as, 2:43, 3:91, 3:97–98, 4:146
 viruses as, 1:182, 1:201, 1:204, 1:222, 2:107
- Vegetative segregation
 defined, 2:197
 in mitochondria and chloroplasts, 2:197
 in plasmids, 2:198
- Venter, Craig, 2:113, 2:173–174, 2:175
- V-erbB* oncogene, 3:130
- Vertical evolution, 1:27
- Vesicles, defined, 1:113
- Vetter, David, 1:77, 4:75, 4:76
- V-fms* oncogene, 3:130
- Viability analysis, 1:187
- Vibrio cholerae*, chromosome shape, 1:139–140
- VIQ (verbal ability) intelligence scores, 2:208, 2:211
- Viral oncogenes (*v-onc*), 4:38
- Viral receptors, 4:166
- Viral tropism, 4:166
- Viral-plasmid vectors, 1:222
- Virions
 characteristics, 4:35
 defined, 4:39
- Viroids, 4:162–163
- Viruses, 4:164–171
 ambisense genomic segments, 4:166
 antisense nucleotide tools, 1:30
 as bioterrorism tools, 1:69
 and cancer, 1:92, 1:100, 3:127, 3:130, 4:34, 4:169–170
 capsids, 4:118–120, 4:164, 4:165
 capsomeres, 4:165
 chaperone proteins and, 4:168
 and chromosomal aberrations, 1:123
 cytotoxic T lymphocytes and, 1:32
 defective particles, 4:119
 defined, 4:162
 and diabetes, 1:210
 distinguished from free-living microbes, 4:164
 dsRNA in, 1:31
 escaped transgene role, 1:68
 genome characteristics, 2:113, 2:117
 helicase primase enzymes, 4:13
 infection outcomes, 4:169
 mutations, consequences, 2:128
 oncoproteins, 4:155
 overlapping genes, 3:135–136
 plant, as biopesticides, 1:57–58
 plants resistant to, 2:108, 2:109, 4:134
 plaques, 4:120
 positive and negative polarity, 4:165, 4:166
 ribozymes, 4:44–45
 size range, 4:164–165
 structures, 4:165–166
 techniques for examining, 4:164–165
 as teratogens, 1:79–80
 as vectors, 1:182, 1:201, 1:204, 1:222, 2:107
 See also Bacteriophages; Retroviruses (RNA viruses)
- Viruses, replication, 4:162, 4:167
 assembly, maturation, and release, 4:168–169
 attachment and penetration, 4:166–167
 budding, 4:168
 infection process, 1:255, 3:126
 transduction, 4:117–120
 uncoating and replication, 4:167–168
- Viruses, specific types of
 adenoviruses, 2:75–78, 3:124, 4:169
 adenoviruses, recombinant, 2:75–77, 2:78
 chickenpox, 2:155
 cytomegalovirus, 3:124
 equine encephalitis, 4:165
 helper, 4:164
 hepadnaviruses, 4:168
 herpes simplex, 2:76, 3:17, 3:124, 4:13
 herpes zoster, 4:165

- Viruses, specific types of (continued)
 herpesviruses, 4:166, 4:169, 4:171
 HPV, 4:155, 4:165, 4:169, 4:171
 HTLV, 4:34, 4:36, 4:38, 4:169
 influenza, 4:165
 Lassa fever, 4:165
 lentiviruses, 2:151
 lysogenic, 4:118–119
 lytic, 4:117–118
 measles, 4:165
 parvovirus, 4:165
 picornaviruses, 4:165
 poliovirus, 4:165, 4:166, 4:168, 4:169, 4:170–171
 poxviruses, 4:164, 4:166, 4:167
 proviruses, 2:152, 4:35–37, 4:39
 reoviruses, 4:167
 rhabdovirus, 4:165, 4:170, 4:171
 rhinovirus, 4:165
 rice dwarf, 4:165
 Rous sarcoma, 1:99–100, 4:39, 4:169
 satellite, 4:164
 Sin Nombre, 4:165
 smallpox, 1:69, 4:165, 4:171
 tobacco mosaic, 4:168
 togaviruses, 4:168–169
 tomato bushy stunt, 4:165
 variola, 4:165
See also Hepatitis viruses;
 HIV/AIDS
- Viruses, treatments
 antibiotics, 1:28
 DNA methylation, 3:47–48
 DNA vaccines, 1:255
 interferon, 1:63
 RNA interference, 4:53–54
 vaccines, 4:170–171
- Virusoids, 4:162, 4:164
 helper viruses, 4:164
 hepatitis delta, 4:164
 satellite, 4:164
 submavirus, 4:164
 subterranean clover mottle virus, 4:164
- Vision
 blindness, 1:7, 1:171–172, 1:179, 1:209, 1:212
 mechanisms, 1:170
 signal transduction role, 4:89
- Vision, color, 1:170–173
 acquired defects, 1:171–172
 hereditary defects, 1:171–172
 inheritance patterns, 3:78
 normal mechanisms, 1:170–171
 opsin protein genes, 1:171
 tests for, 1:172
 as X-linked disorder, 2:203, 3:78
- Vitamins
 A, transgenic rice, 2:106–107, 2:108, 3:149, 4:135
 B12, synthesized by bacteria, 2:10
 C, from genetically engineered bacteria, 1:63
 D3, and hormonal regulation, 2:161, 2:164
 folate, 1:78, 1:128
 K, synthesized by bacteria, 2:10, 2:13
 metabolic role, 3:37, 3:39
- Vitelline membrane, 4:21
- Vitravene, 1:30–31
- VNTRs. *See* Variable number of tandem repeats
- Vogelstein, Bert, 3:131
- Volcanic winter, and population bottlenecks, 3:168
- Volta, Alessandro, 1:37
- Von Willebrand factor, 2:145
- V-sis* oncogene, 3:130
- V-src* (viral-sarcoma) gene, 1:100
- Vulnerable population status, 1:186
- W**
- Waldeyer, Heinrich Wilhelm, 1:132
- Wallace, Alfred Russel, 4:92
- Walleye dermal sarcoma virus, 4:35
- Warren, Stephen, 2:42
- Watermelons, seedless, 3:165, 3:165
- Watson, James, 4:171–173
 Delbrück and, 1:204, 4:172
 DNA structure research, 1:62, 1:192, 1:251, 2:50, 2:120, 2:171, 3:105, 4:172
 on genetic coding, 1:252
 Human Genome Project role, 2:173–174, 2:175, 2:176, 4:173
 Muller and, 4:172
- Web sites. *See* Internet
- Wechsler Adult Intelligence Scale, 2:208
- Weed control, herbicide-resistant crops, 1:58
- Weidemann-Rautenstrauch syndrome, progeroid aspects, 1:2
- Weight gain
 and aging, 1:7
 and cardiovascular disease, 1:101
 and diabetes, 1:210
- Weinberg, Robert, 3:127
- Weinberg, Wilhelm, 2:133
- Weissmann, Charles, 3:188
- Weldon, Walter F. R., 3:102
- Werner's syndrome, 1:1, 1:3
- Western blotting, 1:86
- Wet lab, defined, 1:181
- Whales
 illegal trade concerns, 1:189
 population bottlenecks, 1:189
- What is Life?* (E. Schrödinger), 1:192, 1:203
- Wheat
 genome size, 1:34
 Hessian fly resistance, 1:58
- Wheat, polyploid, 2:114, 3:166
- White, Raymond, 3:130
- Widow's peak hairlines, 2:191, 2:202
- Wiener and Landsteiner, Rh system, 1:84
- Wigler, Michael, 3:127
- Wild-type alleles/traits, defined, 1:45–46, 3:100, 4:137
- Wilkins, Maurice, 4:172
- Wilms tumor
 defined, 2:132
 imprinting disorder, 2:185
- Wilson, Allan, 3:67
- Wilson, Edwin, 3:73
- Wiskott-Aldrich syndrome, 1:68
- Wobble hypothesis, 2:85, 4:4
- Wolffian ducts, 1:21, 1:22, 1:23, 4:79, 4:79
- Wolfram syndrome, 1:212
- Working Group on Ethical, Legal, and Social Implications of the Human Genome Project, 2:94
- World Health Organization, addiction criteria, 1:4
- World Trade Center, terrorist attacks, 1:70
- World War II Veteran Twins Registry, 4:159, 4:160
- Wright, Sewall, 3:103, 4:1
- Wright, Woodruff, 4:105
- X**
- X chromosomes, 2:114, 4:173–176
 aberrations, detecting, 3:183
 characteristics, 4:174, 4:174, 4:175
 and color vision, 1:171–172
 condensation of, 2:63
 crossovers, 1:196–197
 fruit fly, 2:43, 3:7, 3:73, 3:164
 heterochromatin, 1:134–135
 human, 1:137, 2:113, 2:199, 3:106, 3:164
 inactivation of, 2:63, 2:204, 3:76–78, 3:111, 4:175–176
 monosomy, 3:111

- and sex determination, 2:199, 2:203, 3:76, 3:77, 3:106, 4:80–81, 4:174
- and sex determination, nonmammals, 4:82
- sexual orientation role, 4:83–84
- sperm production role, 4:174
- See also Chromosomes, sex; Mosaicism; Trisomy
- X inactive specific transcripts (*XIST*) gene, 3:77
- Xenobiotic, defined, 2:163
- Xenobiotic receptors (SXR), function, 2:163
- Xenografts, 4:139, 4:142–143, 4:142
- Xenopus laevis* (African clawed frog), as model organism, 3:60, 3:61
- X-linked disorders
- AIS, 1:21–26
 - assays for, 3:93
 - dominant, 2:203, 4:175
 - genetic testing, 2:101
 - glucose-6-phosphate dehydrogenase deficiency, 2:148
 - inheritance patterns, 4:173, 4:174–175
 - insertion/deletion mutations, 2:86
 - Leigh's disease, 3:54
 - in males, 3:78
 - ornithine transcarbamylase deficiency, 3:45
 - preimplantation genetic diagnosis, 3:186, 3:187
 - recessive, 1:213, 2:203, 4:174, 4:175
 - Rett syndrome, 2:202, 3:48, 3:49, 3:78
 - SCID-X1, 4:75
 - Simpson-Golabi-Behmel syndrome, 2:132
 - See also Hemophilia; Klinefelter's syndrome; Mosaicism; Muscular dystrophy, Duchenne; Turner's syndrome
- X-linked inheritance
- color vision, 1:171–172
 - fruit fly studies, 1:130–131, 2:43, 3:73–74, 3:91–92, 3:102
- Xq28 gene, sexual orientation role, 4:84
- X-ray crystallography, 4:48, 4:165
- X-ray diffraction, 3:71
- X-rays
- dose-mutability relationships, 3:103
 - as mutagens, 3:81, 3:88, 3:90–92, 3:97, 3:103, 3:166–167, 4:172
- Y**
- Y chromosome Adam, 3:67–68
- Y chromosomes, 4:176–179
- aberrations, detecting, 3:183
 - banding techniques to identify, 1:135
 - characteristics, 4:174, 4:174, 4:175, 4:177
 - DNA repair, 4:177
 - fruit fly, 2:43, 3:164
 - genome characteristics, 3:76
 - human, 2:113, 2:114, 2:199, 3:106, 3:164
 - as indicators of human evolution, 2:167, 2:198, 3:66, 3:168, 4:174, 4:177–179, 4:178
 - inheritance patterns, 4:176–177
 - mutations, consequences, 4:177
 - polymorphisms, 4:178
 - recombination, lack of, 3:65, 4:174, 4:177, 4:178
 - and sex determination, 1:21, 1:23, 2:199, 2:203, 3:65, 3:76, 3:106, 4:80–81, 4:126–127, 4:176–177
 - SRY* gene, 1:21, 3:76, 4:80–82, 4:126–127, 4:177
 - testes development role, 1:21, 1:23
 - X-Y crossovers, 1:196–197
 - See also Chromosomes, sex
- YACs (yeast artificial chromosomes), 1:144–145, 1:222–224
- Yanagimachi, Ryuzo, 1:163
- Yangtze River dolphin, 3:167
- Yeast Proteome Database, 3:209
- Yeast two-hybrid system, 2:9, 3:209, 4:181
- Yeasts (*Saccharomyces* sp.), 4:179–181, 4:180
- aging studies, 1:7
 - artificial chromosomes, 1:144–145, 1:222–224
 - centromere structure, 1:114
 - DNA methylation lacking, 3:46
 - exceptions to universal code, 2:87
 - in fermentation, 1:70
 - fermentation role, 1:66
 - GCN5-regulated genes, 1:138
 - genetically engineered, 1:62, 4:180–181
 - genome characteristics, 1:132, 2:121, 2:172, 4:179–180
 - haploid complement, 2:114, 4:179
 - inteins, 3:181
 - as model organism, 3:60, 4:179–180
 - recombinant DNA tools, 4:171
 - Ssa and Ssb proteins, 1:117
 - SWI* and *SNF* genes, 1:138
- Yellowstone National Park, Archaea adaptations, 1:37, 1:38
- Yersinia pestis*, genome characteristics, 2:121
- Z**
- Zea mays*. See Maize (corn)
- Zebrafish (*Brachydanio danio*), 4:181–183
- developmental processes, 1:205, 4:182–183
 - DNA libraries of mutations, 2:122, 4:182–183
 - genome characteristics, 2:122, 4:182–183
 - as model organism, 3:60, 3:61, 4:182
 - RNA interference, 4:53
- Zimmer, Karl, 3:103
- Zinc, hyperaccumulators of, 1:60
- Zinc-finger DNA-binding motif, 4:114, 4:115
- Zona drilling, 4:22
- Zona pellucida, 2:34, 2:35, 4:21–22
- Zoological parks, conservation genetics applications, 1:187
- Zygotes
- defined, 2:35, 3:109, 4:22
 - as diploid, 2:115, 3:24–29
 - monosomic, 3:109, 3:111
 - mtDNA in, 3:51
 - transfer, in vitro fertilization, 4:22–23
 - trisomic, 3:109, 3:111
- Zymogens (pro-proteins)
- polypeptide cleavage by, 3:179, 3:181